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A review of artificial intelligence techniques for anomaly detection in smart grid

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Abstract
In the era of smart grids (SGs), as more interconnected energy sources and renewable sources are used, it is becoming increasingly important to have robust and accurate advanced anomaly detection methods. Due to the complexity of modern power systems, anomalies need to be detected more efficiently. This study provides a comprehensive overview of integrating renewable energy sources into SGs and the increasing importance of robust anomaly detection methods in ensuring grid security and reliability. Addressing four key research areas, we explore the current trends in applying machine learning techniques to SG anomaly detection research, identifying anomalies such as electricity theft, cyber-attacks, power system disturbances, and abnormal consumption patterns. We systematically evaluate the utilization of different machine learning models, including supervised, unsupervised, semi-supervised, and reinforcement learning, to detect each anomaly within SG environments. Furthermore, we assess the effectiveness of the anomaly detection algorithms and discuss the potential for further research, emphasizing the need for multidisciplinary collaboration and continuous development to overcome challenges and adapt to evolving grid dynamics and cyber threats. The findings of this study suggest that machine learning significantly contributes to ensuring the resilience and efficiency of SGs in the face of evolving challenges.

Keywords Anomaly detection · Artificial intelligence · Machine learning · Power grid · Smart grid

Abbreviations

1D-CNN	1-Dimensional Convolutional Neural Network
AI	Artificial Intelligence
AMI	Advanced Metering Infrastructure
ARIMA	Auto-Regressive Integrated Moving Average

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AUC	Area Under the ROC Curve
CDIA	Covert Data Integrity Assaults
MLP	Multilayer Perceptron
DDoS	Distributed Denial of Service
DER	Distributed Energy Resources
FL	Federated Learning
DT	Decision Trees
EC	Exclusion Criteria
EVs	Electric Vehicles
FDIA	False Data Injection Attacks
DBNs	Deep Belief Networks
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
NIST	National Institute of Standards and Technology
PCA	Principal Component Analysis
PMUs	Phasor Measurement Units
Q1	Quartile 1
Q2	Quartile 2
ACM	Computing Machinery
RF	Random Forests
RL	Reinforcement Learning
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Networks
GAN	Generative Adversarial Network
FCM	Fuzzy C-Means
FN	False Negative
FP	False Positive
FRL	Fuzzy C-Means state prediction Reinforcement
MDPI	Multidisciplinary Digital Publishing Institute
GLAD	Grid Load Anomaly Detection
GNB	Gaussian Naive Bayes
HTM	Hierarchical Temporal Memory
IC	Inclusion Criteria
ICT	Information and Communication Technology
IEEE	Institute of Electrical and Electronics Engineers
IF	Isolation Forest
IoT	Internet of Things
KNN	K-Nearest Neighbors
LSTM	Long Short-Term Memory
DL	Deep Learning
SPE	Specificity
SEN	Sensitivity
SCADA	Supervisory Control and Data Acquisition
XGBoost	Extreme Gradient Boosting
SG	Smart Grid

SLRs	Systematic Literature Reviews
SQ	Search Queries
SSAE	Semi-Supervised AutoEncoder
SVM	Support Vector Machines
TF	Theft Confidence
TN	True Negative
TP	True Positive
V2G	Vehicle-to-Grid
S2G2M2	Semi-supervised Generative Gaussian Mixture Model
SSL	Semi-supervised Learning

1 Introduction

Due to the ongoing industrial revolution, higher electricity costs, and environmental concerns such as global warming, governments and energy providers are actively enhancing the efficiency and reliability of power grids. In addition, they are exploring renewable energy sources and integrating them seamlessly into existing power transmission and distribution systems to meet the growing energy demands of the twenty-first century (Faheem et al. 2018). Renewable energy sources, such as solar, wind, tidal, and biomass generators, have increased dramatically in recent years. The current power grid is a complex network with hundreds of generators using various technologies and setups. Renewable systems, which range in size from a few kilowatts to several megawatts, are replacing traditional energy generators. Managing all of these diverse components digitally leaves the system vulnerable. To ensure the secure and efficient operation of the power grid, it is essential to use smart, two-way communication methods (Sun et al. 2018).

The electrical power system has significantly changed from traditional setups to more advanced smart grids (SGs) over the past decade. Researchers have been paying increasing attention to this change in recent years. In contrast to older models, SGs are designed to provide information and power in both directions, as opposed to relying on a few large, centralized power sources. An extensive network of cables was used to transport power to consumers in older systems, with power generation primarily centralized and dependent on sources such as hydropower or fossil fuels. Despite this, intelligent grids enable a dynamic, two-way flow of power and information resulting from integrating various distributed and renewable energy sources (Ibrahim et al. 2020).

SG technologies are based on the widespread use of information and communication technology (ICT), which facilitates the exchange of data between utility providers and consumers in a two-way manner. An important aspect of advanced metering infrastructure (AMI) is the ability to monitor energy consumption in real-time, negotiate tariffs dynamically, and control load.

Nevertheless, using electric appliances and integrating renewable energy sources into SG systems present new challenges. Electric appliances present a difficulty for the grid since they require the maintenance of system stability and the management of energy distribution locally. The growing need for electric vehicles (EVs) and the implementation of technologies such as dynamic demand control in appliances exert strain on the grid infrastructure. Installing high-power EV chargers can lead to substations experiencing overload concerns

at peak hours (Postan 2024). An anomaly is an unusual event, whether it is irregular consumer behavior, infrastructure issues, power failures, cyber-attacks, or energy theft (Banik et al. 2023a). Detecting these anomalies in SGs is crucial because it helps identify unusual events or behaviors that may indicate problems. The ability to respond quickly to issues like faults in power systems, unusual consumption patterns, power outages, anomalies in load forecasting or smart meter data, cyber-attacks, or energy theft is possible. For the SG to operate efficiently and effectively, timely detection and response are essential for its reliability and security.

In SGs, Machine Learning (ML) and artificial intelligence (AI) can be used to detect anomalies and ensure seamless electricity delivery. A number of anomalies have been identified using ML algorithms, such as random forests (RF), Long Short-Term Memory (LSTMs), and decision trees (DT), including anomalies due to cyber-attacks (Abdelkhalek et al. 2022; Al-Tamimi et al. 2024; Sinha et al. 2024) (Elmrabit et al 2020) and electricity theft (Ali et al. 2022). In addition to detecting irregular consumption patterns (Fenza et al. 2019), these algorithms also detect anomalies in smart meter data (Jaiswal et al. 2020). By leveraging Supervised Learning (SL), Unsupervised Learning (UL), Semi-Supervised Learning (SSL) and Reinforcement Learning (RL) techniques, ML models enhance the security of SGs by identifying and mitigating threats, such as False Data Injection Attacks (FDIA) (Ashrafuzzaman et al. 2020), and Distributed Denial of Service (DDoS) attacks (Diaba and Elmusrati 2023). Additionally, ML assists in predictive maintenance, forecasting faults (Helong et al. 2019), and ensuring grid stability (Matijašević et al. 2022). Integrating AI-driven anomaly detection systems enables real-time monitoring, resulting in quick responses to anomalies (Hyder et al. 2023; Al-Haija et al. 2024). These technologies collectively contribute to the resilience of SGs, safeguarding against disruptions and ensuring a reliable and continuous electricity supply.

This research highlighted the importance of ML algorithms for detecting anomalies in SGs. It assesses various strategies for finding a wide range of anomalies. Providing valuable information about state-of-the-art technology empowers researchers with the necessary knowledge to develop robust anomaly detection solutions, thus improving security and dependability of SGs.

The following are the major contributions of this review paper.

- An in-depth analysis has been conducted comparing traditional and SGs, exploring features, benefits, stakeholders, challenges, AI integration, and anomaly detection.
- The background of SGs is examined, focusing on anomaly types, datasets, ML model types, and algorithms.
- A comprehensive research methodology is presented, including data sources, search queries, inclusion/exclusion criteria, and quality assessment.
- A brief literature on SGs is presented, emphasizing cyber-attacks, electricity theft, energy fraud, data anomalies, physical faults, and predictive maintenance.
- Four critical questions in the domain of SG have been answered, highlighting current trends in ML for SG anomaly detection, types of anomalies detected, ML models employed, and evaluating results and potential scopes for further research.

Figure 1 depicts the structure of the paper. The remainder of the article is arranged in the following manner: Section II discusses SGs, their features and benefits, stakeholder groups,

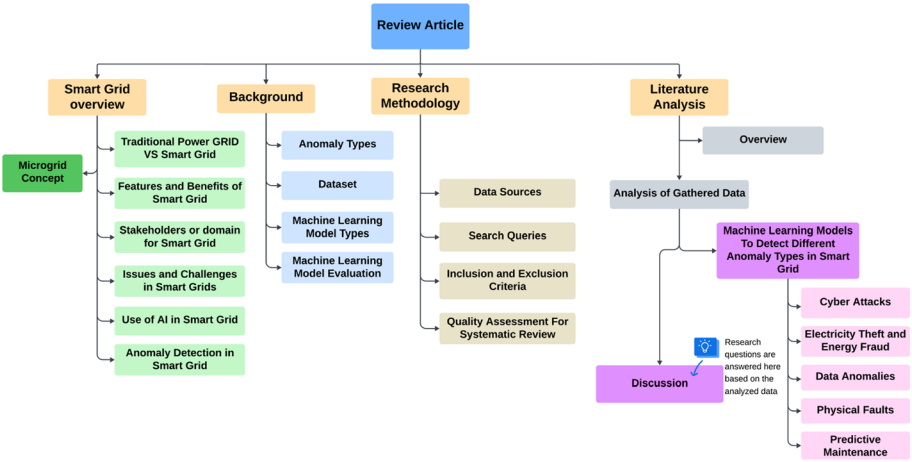


Fig. 1 Structure of the paper

issues, challenges, and the concept of microgrids and AI in SGs. In Section IV, the article selection process, sources, and data assembly processes are described. A systematic review of articles is conducted in Section IV to assess their quality. Finally, the data is analyzed based on anomaly types, ML algorithm types, experiment outcome, and methodology, followed by Sections V and VI addressing research areas, and Section VII with a conclusion.

2 Smart grid overview

In this section, a concise introduction to the SGs is provided, followed by a brief analysis of the Microgrid Concept. Subsequently, an in-depth examination of the stakeholders or domains involved in SGs is presented. Following this, a thorough review of the issues and challenges encountered in SGs is discussed. The incorporation of AI in the SGs is then explored, and finally, a brief overview of Anomaly Detection in the SGs concludes the section.

2.1 Smart grid

An SG is a sophisticated and integrated electricity network incorporating advanced technology, bidirectional communication, and computational intelligence to create a more efficient, reliable, and sustainable energy delivery system. In addition to integrating renewable energy sources, it enhances grid security and resilience, as well as meeting the evolving needs of the energy sector. SG can also be called an "intelligent" network since it helps balance power production, distributes energy sources, and reduces problems such as overloads and voltage fluctuations (Vijayapriya and Kothari 2011). Table 1 compares many aspects of traditional power grids and SG, emphasizing the transition from primarily electromechanical systems to digital technology.

Bidirectional real-time communication, distributed power generation, a dispersed network topology, and substantial sensor use characterize SG. These features enable improved

Table 1 A brief comparison between traditional power grids and the SGs

Aspect	Traditional grid	Smart grid	References
Mechanization	Primarily Electromechanical	Digital	Mei and Zhu (2013)
Communication	One-way communication	Two-way real-time communication	Gao et al. (2012)
Power Generation	Centralized power generation	Distributed power generation	Dibangoye et al. (2015), Jarrah et al. (2015)
Network Structure	Radial Network	Dispersed Network	Bianco et al. (2015)
Data Volume	Involves less data	Involves large volumes of data	Macedo et al. (2015)
Sensors and Monitors	Relies on a small number of sensors	Utilizes many sensors and monitors	Kilic and Gungor (2013), Wang et al. (2015)
Monitoring	Less or no automatic monitoring	Great automatic monitoring	Sendin et al. (2013), Grzonka et al. (2015)
Control and Recovery	Manual control and recovery	Automatic control and recovery	Dahal et al. (2015)
Security and Privacy	Fewer security and privacy concerns	Prone to security and privacy issues	Kessler et al. (2015)
System Disruptions	Requires human attention	Utilizes adaptive protection	Mirsaeidi et al. (2014)
Energy Consumption and Storage	Simultaneous production and consumption of energy/electricity	Incorporates storage systems	Zaghib et al. (2015)
Control System	Limited control	Extensive control system	Kim et al. (2015), Evora et al. (2015), Ahn and Peng (2013), Järventausta et al. (2010)
Emergency Response	Slow response to emergencies	Fast response to emergencies	Tuballa and Abundo (2016)
User Choices	Offers fewer user choices	Provides vast user choices	Farhangi (2009)

monitoring, automatic control and recovery, and more significant data volumes. While providing extensive options for users, SG is prone to security and privacy issues. However, they outperform in their ability to protect against disruptions adaptively, integrate storage systems, and facilitate rapid reactions to emergencies, enhancing their efficiency, reliability, and environmental sustainability compared to traditional grids. The advantages of the SG encompass higher energy efficiency, diminished environmental footprint, expedited emergency response, fortified security measures, consumer empowerment, optimized energy allocation, and integration of nascent technologies such as EVs. SG increases energy efficiency by using renewable energy, demand response, and distributed control technology. These characteristics help SG optimize power supply and usage, improving efficiency. Photovoltaic power is efficiently used and incorporated into the grid, improving energy generation and consumption (Anjana and Shaji 2018; Bakare et al. 2023**).** As a result of the modernization of power systems, SGs and innovative networks have emerged, revolutionizing electricity generation, distribution, and consumption. Table 2 provides an overview of the SGs' key features and benefits, emphasizing their transformative impact on traditional electrical infrastructure.

There are numerous advantages to SGs, including the ability to enable bidirectional information and electricity flow (Fang et al. 2012) and the ability to promote sustainability (Gharavi and Ghafurian 2011a; Wang et al. 2020) by integrating renewable energy sources. Additionally, there are market-enabling mechanisms (Tuballa and Abundo 2016) that enable users to participate actively in the energy ecosystem through enhanced efficiency (Tuballa and Abundo 2016; Gharavi and Ghafurian 2011a; Abosedra et al. 2009) and resilience in the face of disruptions. Considering these advantages, we can then explore the various aspects of SGs that make them an essential advancement in power distribution. Figure 2 illustrates the major features of SGs.

2.2 Issues and challenges in smart grids

SGs introduce many complexities as the world transitions toward more sustainable and technologically advanced energy systems. Providing a concise overview of the diverse challenges associated with SGs' implementation, Table 3 provides a brief overview of security concerns in complex networks as well as the need for interoperability and standards. The challenges range from technical aspects such as routing design (Saputro et al. 2012) and data protection (Rawat and Bajracharya 2015; Bekara 2014) to broader issues such as cost–benefit analysis and consumer engagement (Crispim et al. 2014; Xenias et al. 2015). Table 3 concisely overviews the primary obstacles encountered in implementing and securing SG systems, covering security risks in complex networks, vulnerabilities in smart meters, and outdated equipment.

The technical difficulties encompass optimizing data flow, ensuring interoperability, and enhancing cybersecurity measures. Additional challenges involve improving SG simulators, developing context-aware middleware platforms, guaranteeing compatibility with intelligent devices, integrating distributed generation, and implementing customized data encryption and security solutions.

Table 2 Features and benefits of SGs

Features or benefits	Description	References
A two-way flow of electricity and information	SGs are unique in that they enable two-way information and electricity flows. As opposed to traditional power grids that transmit power primarily from centralized generators to consumers, an SG facilitates the exchange of electricity and data in multiple directions. As a result of this bidirectional flow, the entire grid can be controlled, monitored, and optimized more effectively	Fang et al. (2012)
Integration of technologies	Information technology, automation, telecommunications, and control systems are all integrated into the SGs. Smart meters, sensors, digital network management devices, and bidirectional communication are among the components that enable the processing and decision-making of real-time data	Gharavi and Ghafurian (2011a), Greer, et al. (2014), Abosedra et al. (2009)
A cyber-secure communication system	Cyber-secure communication technologies are crucial to protecting data and infrastructure in SGs. This ensures reliability and safeguards the grid against malicious attacks	Gharavi and Ghafurian (2011a), Greer, et al. (2014), Gharavi and Ghafurian (2011b)
Improved efficiency	With SGs, power delivery can be optimized through improved demand-side management, energy efficiency, and self-healing capabilities, thus reducing waste and contributing to a more sustainable energy future	Tuballa and Abundo (2016), Gharavi and Ghafurian (2011a), Abosedra et al. (2009)
Resilience	In order to maintain a stable power supply even during adverse conditions, SGs can adopt strategies that can respond to events anywhere in the grid, whether caused by natural disasters or sabotage	Gharavi and Ghafurian (2011a), Wang et al. (2020)
Sustainability	By integrating renewable energy sources into the grid, SGs enable high penetration of clean and intermittent energy resources, thus reducing pollution and environmental impacts	Gharavi and Ghafurian (2011a), Wang et al. (2020)
Market Enabling	SGs enable efficient energy use and economic welfare by allowing consumers to consume energy and sell excess energy back to the grid	Tuballa and Abundo (2016)
Integration of Users	By providing consumers with the option to consume energy and sell excess energy back to the grid, SGs enable efficient energy use and economic welfare	Ardito et al. (2013)

2.3 AI in smart grid

In the SGs context, AI has also proven to be highly beneficial. SGs require high levels of stability, security, and reliability, all of which are effectively supported by AI techniques. As a result of these AI applications, including ML, deep learning (DL), and RL, critical aspects of the SG have been enhanced, including security assessment, stability assessment, fault diagnosis, and stability control (Marugán et al. 2018; Vázquez-Canteli and Nagy 2019). It is important to note that AI has been shown to be capable of improving the accuracy, speed, and effectiveness of stability analysis and control processes while simultaneously reducing human workloads (Shi et al. 2020). As a result, these AI-driven studies have achieved notable results in ensuring the stability and reliability of SGs (Tables 4, 5).

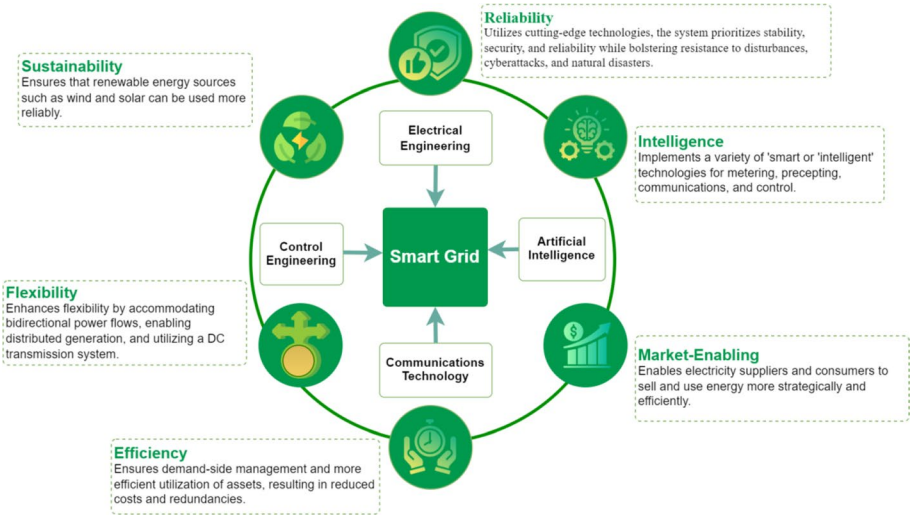


Fig. 2 Major features of SGs

AI applications in SGs face many practical challenges, including the need for extensive data, imbalanced learning (the distribution of classes in the dataset is highly skewed), interpretability of AI models (explanation of the black box nature of the AI models), challenges with transfer learning, robustness to communication quality issues such as noise, data loss, and time delay, as well as the ability to withstand attacks and adversarial examples (Shi et al. 2020). AI's potential in the SG can only be fully realized if these challenges are addressed.

2.4 Anomaly detection in smart grid

An anomaly in the SG context denotes any abnormal event, stemming from irregular consumption, infrastructure failures, cyber threats, or energy theft (Banik et al. 2023). Example of anomalies falling within this category are: electricity theft (Ali et al. 2022; Eddin et al. 2022), cyberattacks (Elmrabit et al. 2020, Hyder et al. 2023), FDIA (Esmalifalak et al. 2017; Zia et al. 2023), DDoS (Diaba and Elmusrati 2023), natural disturbances (Panthi et al. 2020), operational anomalies (Siniosoglou et al. 2021), non-technical losses (NTL) such as faults or electricity theft (Lu et al. 2019), irregularities or fraud in electricity consumption (Oprea and Bâra 2021), grid load anomalies (Wei et al 2020), power anomalies (Hollingsworth et al. 2018), and anomalies in smart meter data (Sun et al. 2022; Jaiswal et al. 2021). It is essential to detect and address these anomalies to maintain a reliable and secure electricity supply.

Anomalies, especially those pointing to cyberattacks, may pose serious risks to SG systems if someone gains unauthorized access or alters data. Cybersecurity is of paramount concern due to the increasing number of cyber threats and security incidents targeting electricity infrastructure (Yang et al. 2011).

Besides intentional attacks, there are also unintentional compromises, such as user errors or equipment failures, that can pose problems. A malicious attack could exploit these vulnerabilities to gain access to edge devices (Sinha et al. 2024) or to control centers and mod-

Table 3 Issues and challenges in the SGs

Issue/challenge	Description	References
SG security	As SG networks play a critical role in energy distribution, their security is imperative. The challenges include vulnerabilities in smart meters, the prevalence of outdated equipment, and the inherent risks associated with device-to-device communication. Furthermore, maintaining robust security protocols is complicated by the diversity of stakeholders involved. To address these challenges, a multifaceted approach is required, which includes the latest technological safeguards, continuous system updates, and stakeholder collaboration	Bekara (2014), Wang et al. (2011), McBride and McGee (2012), Farag et al. (2014)
V2G (Vehicle-to-Grid)	V2G technology, which allows EVs to return energy to the power grid, faces a number of significant challenges. These concerns include battery wear and tear, the low penetration rate of EVs in the market, and the need for innovative battery technologies that can endure frequent charging and discharging cycles. The successful implementation of V2G technology, which offers enhanced grid stability and the integration of renewable energy, will depend on overcoming these challenges	Mwasilu et al. (2014)
Implementation barriers	SG technologies face several barriers, including conducting comprehensive cost–benefit analyses, bridging knowledge gaps among stakeholders, and overcoming institutional inertia. Due to these barriers, a swift transition to SG systems is hindered, emphasizing the need for targeted education programs, policy incentives, and a clear demonstration of the advantages of SG systems in the long run	Muench et al. (2014)
Costs and consumer engagement	There is a significant cost associated with implementing SG technologies, and consumer engagement is crucial to their success. There is a need to develop strategies to decrease these costs effectively while promoting consumer participation through education, incentives, and open and transparent communication about SG benefits. To achieve the energy efficiency and environmental goals associated with SG technology, it is crucial to have a dual focus	Crispim et al. (2014), Xenias et al. (2015)
Demand Response Optimization	Demand response is important for improving SGs' flexibility by adjusting consumer demand to real-time signals or dynamic pricing. However, implementing optimized SGs could be challenging as it must account for heterogeneous consumer devices, stochastic consumption patterns, and latency in communication networks. In addition, ensuring the pricing range aligns with real-time operational constraints requires accurate load forecasting, fast-response algorithms, and secure, low-latency communication infrastructure	Sharma and Tiwari (2024), Sinha et al. (2022)
Data protection and privacy	Data protection and privacy are significant concerns associated with the transition to SGs. An SG system generates and transmits a substantial amount of data, which poses the threat of unauthorized access and misuse. It is imperative to safeguard consumer information by implementing stringent data protection measures, implementing privacy-enhancing technologies, and complying with all regulatory requirements	Rawat and Bajracharya (2015), Bekara (2014)
Routing design	SGs' optimal operation depends on efficient data routing and communication. It is challenging to design a routing system that can handle the high volume and speed of data transfer while maintaining reliability and minimizing delays. The development of routing protocols that are able to adapt to the dynamic nature of SGs requires innovative approaches and technologies	Saputro et al. (2012)

Table 3 (continued)

Issue/challenge	Description	References
Interoperability and standards	Interoperability and compliance with standards across diverse components of SG systems can be challenging to maintain. It is therefore necessary to develop universal standards and protocols that will ensure seamless communication and integration of different technologies within the SG ecosystem. Achieving interoperability is crucial to the scalability and efficiency of SGs	Gaviano et al. (2012)
Physical and cybersecurity	In addition to cyber threats, SGs are at risk from physical threats as well. The AMI and other components of an SG are vulnerable to physical tampering as well as cyber-attacks. Enhanced physical security measures are essential for protecting infrastructure and ensuring power grid reliability, alongside cybersecurity practices	Rawat and Bajracharya (2015)
SG simulators	Simulators for SGs are essential for testing and planning. There are, however, a number of limitations to current simulators, such as their inability to reproduce complex scenarios accurately, issues with time synchronization, and insufficient data exchange mechanisms. To overcome these limitations, simulation technology must be advanced, enabling more realistic and comprehensive testing of SG configurations	Li and Zhang (2014)
Context-aware middleware platforms	An SG requires middleware platforms that are scalable, interoperable, and context-aware. It is essential that these platforms are capable of processing and analyzing data from a variety of sources, adapting to changes in conditions, and facilitating communication between different components. The development of such platforms is crucial to ensuring the flexibility and efficiency of SGs	Donohoe et al. (2015)
DERs integration	Applications for DERs such as solar panels, batteries, and EVs are in high demand and offer valuable local energy generation and storage capabilities. However, the bidirectional flow of power introduced by DERs affects voltage control and protection systems. The output from these resources is also variable and difficult to predict. Additionally, different devices often utilize varying communication standards, which complicates coordination efforts. The increasing number of connected devices also raises the risk of cyberattacks. Thus, the grid requires improved sensors, forecasting tools, and control systems to manage these complexities	Abdelkhalek et al. (2022), Sun et al. (2022)
Compatibility with intelligent devices	Compatibility between SG devices is a significant challenge. It requires the establishment of technological standards for logging and communicating data. To ensure seamless integration of devices and overall SG functionality, compatibility issues must be addressed	Muttaqi et al. (2014)
Distributed generation and automation	Automating distribution processes and integrating distributed generation sources present complex challenges. It is important to address technical, regulatory, and operational issues to ensure that distributed generation and automation are utilized to their full potential in improving grid resilience and efficiency	Muttaqi et al. (2015)
Data encryption and security solutions	To protect the SGs against cyber threats, it is necessary to implement robust data encryption and develop customized security solutions. Taking these measures is essential for protecting sensitive data and ensuring its integrity and confidentiality	Wang and Lu (2013)

ify load conditions to destabilize an SG, resulting in significant problems such as brownouts or even catastrophic blackouts. Also, this may cause unexpected economic losses and customer privacy breaches (Yang et al. 2011). Power distribution problems can arise when the grid has inaccurate or anomalous data. Anomalies can significantly impact the accuracy of

Table 4 SG anomaly types

Anomaly type	Description	References
Electricity theft	There has been considerable research devoted to developing methods and models for detecting irregularities related to electricity theft in order to distinguish between normal and suspicious consumers	Eddin et al. (2022), Wang et al. (2019), Zhang et al. (2018), Haq et al. (2023), Javaid et al. (2021), Zafar et al. (2023), Zidi et al. (2023), Hasan et al. (2019)
Cyber-attacks and system intrusions	A cyber-attack on an SG occurs when someone intentionally and maliciously attempts to undermine its confidentiality, integrity, or availability. These attacks can include accessing critical infrastructure components without authorization, manipulating data, or modifying control systems. There are several studies that provide valuable insight into the challenges of detecting anomalies in SGs caused by cyberattacks, as well as innovative approaches to resolving these challenges	Elmrabit et al. (2020), Diaba and Elmusrati (2023), Hyder et al. (2023),FZia et al. (2023), Karimipour et al. (2019), Kosek et al. (2016), Radoglou-Grammatikis and Sarigiannidis (2018), Ünal et al. (2021), Martinelli et al. (2022), Khan et al. (2021)
Power system disturbances	SGs are susceptible to disturbances caused by both natural and man-made events and behave differently due to external factors or malicious interventions. To identify the specific types and causes of power system disturbances, several ML models have been developed and proposed	Jaiswal et al. (2020), Helong et al. (2019), Siniosoglou et al. (2021), Wei et al. (2020), Hollingsworthet al. (2018), Danilczyk et al. (2020), Jithish et al. (2023), Moghaddass and Wang (2018), Li et al. (2023)
Abnormal power consumption patterns	Research concerning abnormal power consumption patterns involves deviations from expected electricity consumption behavior, and the researchers aim to create models that can detect abnormal patterns, whether they result from faults, anomalies, or fraud	Fenza et al. (2019), Jaiswal et al. (2020), Matijašević et al. (2022), Dai et al. (2020), Yuan et al. (2015)

Table 5 Datasets used in various studies

Publicly available	KDDCUP 99; NSL-KDD; ISCX-IDS2012; CIC-IDS2017; CSE-CIC-IDS2018; Bot-IoT; IoT-23; CICIDS-2017; UNSW-NB15; ICS cyberattack; OpenEI; NPCS; ElectricityLoadDiagrams201120145; Smart Meter Dataset; SCADA; SGCC; ISO New England (NEPOOL); TDD2022; Meteorological data; SG Dataset; NYISO; Commercial and Residential Hourly Load Profiles; DNP3 intrusion detection; HAI Security Dataset
Available on request	μPMU dataset; Smart Meter Consumption Data collected from University of Massachusetts; Irish CER; Transformers collected data; ISET
Simulation/generated	IEEE 6-bus; IEEE 14-bus; IEEE 118-bus; IEEE 3-bus; Microgrid simulation; IEEE 39-bus; IEEE 57-bus; IEEE 18-bus; IEEE 9-bus; IEEE 300-bus; IEEE 2848-bus

predictive models for load forecasting and demand planning (Cui et al. 2019). If predictions are inaccurate, managing the grid and allocating the resources may be difficult.

An anomaly detection technique, also referred to as an outlier detection technique, is used to identify unusual instances in a dataset that differ significantly from the typical data (Banik et al. 2023). By applying AI technologies to various data sources such as networks, IoT devices, SCADA systems, SGs, smart meters, and machine logs, data can be categorized into two categories: (1) normal behavior, and (2) abnormal behavior. AI can perform binary classification of normal and abnormal behavior, or multi-class classification to identify abnormal behaviors associated with specific attacks or predict irregularities, attacks, or events (Altamimi and Abu Al-Haija 2024) (Al-Tamimi et al. 2024; Elmrabit et al. 2020). ML algorithms bring significant benefits in identifying these anomalies by using patterns and deviations from the norm. This enhances the ability to detect irregularities or unusual events within complex datasets efficiently. A number of proposals in the literature used ML-based approaches to address the challenges of anomaly detection.

It is possible to identify different types of anomalies using various anomaly detection algorithms, including SL, UL, SSL, RL, etc. To detect cyber-attacks and intrusions, researchers have employed algorithms such as Support Vector Machines (SVM) (Esmalifalak et al. 2017), RF (Elmrabit et al. 2020), Hierarchical Temporal Memory (HTM) (Barua et al. 2020), Naive Bayes (NB) classifiers (Abdelkhalek et al. 2022), and distributed SVMs (Esmalifalak et al. 2017). The isolation forest (IF) algorithm is also a promising approach to detecting Covert Data Integrity Assaults (CDIA) in SG communication networks (Ahmed et al. 2019). The detection of electricity theft involves the application of various algorithms, including fine tree regression models (Ali et al. 2022), Extreme Gradient Boosting (XGBoost) classifiers (Buzau et al. 2019), density-based spatial clustering, FCM clustering (Qi et al. 2022), kernel density estimation, one-class SVMs, autoencoders, Principal Component Analysis (PCA) (Ali et al. 2022), cubic SVMs (Hashatsi 2020), and LSTM networks (Madhure et al. 2020). Using a Convolutional Neural Network (CNN) architecture, faults in power systems can be identified (Danilczyk et al. 2020). Furthermore, the distributed

SVMs (Esmalifalak et al. 2017) and LSTM networks (Fenza et al. 2019) may be used to detect stealth attacks within power networks. Several artificial neural networks are used in energy fraud detection, including fine-tuned RNN detection algorithms (Eddin et al. 2022) and cubic SVM classification algorithms (Hashatsi 2020). Recently, deep SSL techniques have been used in electricity theft detection and SG cyber-attack detection (Qi et al. 2023, 2021). RL-based approaches have also been explored for online cyber-attack detection in SG using model-free RL frameworks (Li et al. 2023), anomaly detection in smart meter data using Deep RL with FCM forecasting (Siniosoglou et al. 2021), and demand response optimization under dynamic pricing and user preferences (Donohoe et al. 2015). As part of relay protection in smart power grids, SVM classification models have been developed (Dai et al. 2020) for anomaly detection. Furthermore, NB, SVMs, and Logistic Regression algorithms are used to detect fault types in smart energy meters (Helong et al. 2019). The algorithms are further classified based on their application and model types in Table 6.

Table 6 ML model types

ML model type	Description	Usage	Algorithms	References
Supervised	Models are trained on labeled data to make predictions on new data	Used for classifying electricity theft, cyber-attacks, power anomalies, and stability predictions	Linear Regression, RF, XGBoost, Light Gradient Boosting (LGB), GNB, DT, SVMs, KNN, RNN, LSTM, CNN	Jaiswal et al. (2020)
Unsupervised	Utilizes unlabeled data to detect patterns, structures, or anomalies	Employed to detect anomalies, anticipate power usage, and identify aberrant data in smart meters	K-Means Clustering, IF, RNN, LSTM, CNN, HTM, Auto-encoder Neural Network	Fenza et al. (2019), Hollingsworth et al. (2018), Barua et al. (2020), Ahmed et al. (2019), Karimipour et al. (2019), Moghaddass and Wang (2018), Yuan and Jia (2015), Ford et al. (2014), Shabad et al. (2021), Takiddin et al. (2022), Zhang et al. (2019)
Semi-supervised	Integrates both labeled and unlabeled data to train algorithms	Used to identify anomalies in smart meter readings	Semi-supervised generative Gaussian mixture model (S2G2M2), Semi-Supervised Auto-encoder (SSAE), Convolutional Auto-encoder	Lu et al. (2019), Zhang et al. (2018)
Reinforcement Learning	An agent learns from interactions with its environment to achieve goals through a reward-based system	Used for detecting abnormal data in smart meter readings	Model-free RL, FCM state prediction RL network model (FRL)	Esmalifalak et al. (2017), Sun et al. (2022), Kurt et al. (2019), Mukherjee et al. (2022)

3 Background

The introduction of SG technology has revolutionized power grid infrastructure, increasing its reliability, efficiency, and sustainability by introducing advanced monitoring and control mechanisms. However, as complex systems are connected to vast amounts of data, the SG environment becomes susceptible to various anomalies that can compromise its operational integrity. In recent years, detecting anomalies in SGs, which aim to identify irregularities or unexpected behavior, has become a key research focus.

3.1 Anomaly types

Several studies examine the various aspects of anomaly detection within SGs. The identified anomaly types span a range of issues, including electricity theft, fraudulent behavior in electricity consumption, cyber-attacks, disturbances in power systems, and abnormal power consumption patterns. Table 4 summarizes the different types of anomalies.

From Table 4, it is visible that utility providers handle the issue of electricity theft, which is a significant concern, by conducting research that aims to differentiate between regular and questionable consumer behaviors. The focus is on cyber-attacks and system intrusions, with research emphasizing novel methods for identifying abnormalities resulting from deliberate manipulation of SG components. The paper examines power system disruptions, which can arise from natural or human-induced factors, and proposes ML methods to detect and classify their underlying causes accurately. The analysis focuses on atypical power usage patterns, which involve deviations from anticipated behavior. Understanding and categorizing these anomaly types is crucial for designing effective anomaly detection systems in SGs.

3.2 Dataset

Datasets used in ML-based anomaly detection in SGs are crucial for developing and evaluating models. Anomaly detection algorithms can be trained, tested, and validated using various public, open-source, and simulation data regarding SGs, smart meters, other power system components, customer invoices, etc. For training models to detect irregularities in electricity consumption, particularly issues like fraud, large datasets containing consumer and invoice data are invaluable (Oprea and Bâra 2021). A power system attack detection model, utilizing ML algorithms, utilizes data and records obtained from phasor measurement units (PMUs) to provide precise and synchronized measurements of the power system's state. This enables the identification of potential cyberattacks and disturbances (Danilczyk et al. 2020). The smart meter is a fundamental component of an SG, generating data that can be used to analyze energy consumption behavior (Jaiswal et al. 2020; Matijašević et al. 2022). Several studies have focused on using ML and DL approaches to detect and analyze thefts from smart meters (Eddin et al. 2022; Ford et al. 2014). Also, time series data related to energy consumption can be analyzed using large datasets of smart meter readings (Oprea et al. 2021). In addition to forecasting and labeling anomalies, electricity distributors' power consumption data can be utilized to propose efficient ML-based classification methods (Jaiswal et al. 2021). By utilizing various ML algorithms, datasets from public or open dataset repositories can be used to evaluate the stability of SGs (Bashir et al. 2021). Researchers

leverage large datasets to address specific challenges related to anomaly detection, such as electricity theft, cyber-attacks, power system disturbances, and abnormal power consumption patterns in SGs. Table 5 includes widely used datasets in SGs, electricity theft, and intrusion detection. It is worth noting that some of the datasets contained self-generated and simulated attacks, and for a wider range of applicability and ML-model compatibility, this review includes all evaluations (see Table 7 in Sect. 5.2 for more details).

3.3 Machine learning model types

SG anomaly detection involves using various model types to develop effective solutions for identifying irregularities and securing grid infrastructure. A brief overview of the multiple types of modeling, such as SL, UL, SSL, and RL, is presented in Table 6.

The choice of these model types is driven by the problems to be addressed, the available labeled data, and the specific requirements of anomaly detection in SGs. The performance of SL is better when labeled data are abundant, whereas UL is better when there is a shortage of labeled data. SSL strikes a balance, while RL is used when a model learns based on interactions with the environment. It is essential that the right type of model be selected for SG anomaly detection systems.

3.4 Machine learning model evaluation

In ML, model performance evaluation is essential in determining whether a model effectively solves a particular problem. According to the problem's nature, selecting the appropriate evaluation metrics for classification or regression tasks is important.

3.4.1 Classification

In classification, models are trained on labeled datasets, making predictions on new data and categorizing it according to predefined categories. Metrics such as accuracy, precision, recall, and F1-score are critical for assessing a model's ability to identify anomalies, especially those related to electricity theft (Wang et al. 2019), cyber-attacks (Elmrabit et al. 2020; Hyder et al. 2023; Mohammed et al. 2024), or power system disturbances (Jithish et al. 2023; Li et al. 2023). An example would be determining whether a smart meter reading represents a normal or anomalous behavior (Zhang et al. 2019).

3.4.2 Regression

Using regression, continuous numerical values can be predicted. To assess the accuracy of regression models in predicting values, metrics such as MAE, MSE, and RMSE, which provide a quantitative measure of the model's performance, are used. Different types of regression models can be employed within SGs to evaluate system stability (Moghaddass and Wang 2018), forecast power consumption levels (Jaiswal et al. 2020), or estimate non-technical losses (electricity theft) (Madhure et al. 2020). An example might be predicting the power consumption for a particular period based on historical data (Jaiswal et al. 2020).

The performance and effectiveness of ML-based anomaly detection models are evaluated using a variety of metrics. These metrics contribute to a comprehensive assessment

of anomaly detection models in SGs, including aspects such as precision, recall, prediction confidence, and overall accuracy. Researchers combine these metrics to ensure that the proposed models address the specific challenges associated with electricity theft, cyber-attacks, power system disturbances, and abnormal consumption patterns.

4 Research methodology

In this section, we focused on exploring studies related to SGs and smart meters, primarily focusing on anomaly detection. Our objective was to conduct a review of the current trends and interests in research on this topic. Another primary objective was to determine where academics are heading in this field and to identify potential research areas. In our review, we examined the use of ML techniques for detecting anomalies in SGs.

4.1 Data sources

Our comprehensive research explored seven distinct databases, each offering resources. As a search engine, Google Scholar provided a broad range of scholarly articles, serving as our initial point of inquiry. After this, we explored the repositories of IEEE Xplore, Science Direct-Elsevier, Multidisciplinary Digital Publishing Institute (MDPI), Springer, Association for Computing Machinery (ACM), Charles Darwin University Library, and Wiley Online Library to enhance the collection of our papers. We conducted our most recent exploration of these databases on July 5, 2025, to ensure that our research stayed current and knowledgeable. We have selected 78 papers from the last twelve years (2013–2025) and selected papers based on our keywords, such as "smart grid," along with terms like "anomaly detection" or "fault detection" to narrow down our search. We also included words like "data analytics," "artificial intelligence," and "machine learning" to find studies that use modern methods to improve SGs.

4.2 Quality assessment for systematic review

To ensure the reliability of the papers included in our systematic review, we checked the quartile of the papers. For our reviews, we selected the papers that fall within the quartiles Q1 and Q2, with a specific emphasis on good impact factor rankings. A visual representation of this distribution is provided in Fig. 3, in which 36 papers were published in Q1 journals, 7 were published in Q2, and 32 were published in conferences.

5 Literature analysis

This section is outlined as three subsections. A brief analysis of the selected papers has been given.

Table 7 Results-based analysis of the proposed solution, incorporating models from SL, UL, SSL, FL, and RL. The Dataset Used column includes data from public (pub), private/simulated/custom (pri), and available-on-request (req) sources

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S	TF	Other
										TP, TN	FP, FN	
1	Abdelkhalek et al. (2022)	Stealthy IT and OT attacks	KDDCUP 99 (pub), NSL-KDD (pub), ISCXIDS2012, CIC-IDS2017 (pub), CSE-CIC-IDS2018 (pub), Bot-IoT (pub), IoT-23 (pub)	ANN (SL)	98.4	99.57	98.02	98.79	–	–	0.28 (FP) 1.32 (FN)	–
2	Elmrabit et al. (2020)	Cyber-Attack	CICIDS-2017 (pub), UNSW-NB15 (pub) and the Industrial Control System (ICS) cyber-attack dataset (pub)	RF (SL)	73.6 (UNSW) 99.9 (CICIDS) 92.4 (ICS) –	72.6 (UNSW) 99.9 (CICIDS) 92.4 (ICS)	73.6 (UNSW) 99.9 (CICIDS) 92.4 (ICS)	69.5 (UNSW) 99.9 (CICIDS) 92.4 (ICS)	–	–	–	–
3	Ali et al. (2022)	Electricity theft, Cyber-Attacks	OpenEI (pub), Nevada Power Clark Station (NPCS pub)	Fine Tree Regression Model (SL)	–	–	–	–	–	>90 (TC)	–	–
4	Fenza et al. 2019)	Fraud, concept-drift, anomalies in energy consumption	Electricity-LoadDiagrams201120145 (pub)	LSTM (UL)	–	78	88	–	–	–	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
5	Jaiswal et al. (2020)	Smart Meter Data—power consumption	Smart Meter Dataset from UCI ML repository (pub)	Linear Regression (SL)	–	–	–	–	–	–	–	49.9 (RMSE)
6	Ashrafuzzaman et al. (2020)	Stealthy false data injection attacks	Simulated on IEEE 14-bus system with MATPOWER (pri)	Ensemble SVM (SL)	89.61	99.93	–	84.72	0.8675	–	0.03 (FP)	99.97 (SPE)
7	Diaba and Elmusrati (2023)	Cyber-Attack: DDoS	CICIDS2017 (pub)	ConvGRU (SL)	99.7	98.1	–	98.9	–	99.9 (DR)	–	–
8	Helong et al. (2019)	Smart meter fault prediction	Real failure data in Chongqing (2015-2018) (pri)	NB (SL)	98	98	98	98	–	–	–	–
9	Matijašević et al. 2022)	Consumption forecasting losses	Sourced from smart meters in Croatian distribution network (pri)	GRU (SL)	–	–	–	–	–	–	–	0.00945 (MSE)
10	Hyder et al. (2023)	Cyber-Attack	Simulated at Pacific Northwest National Laboratory (pri)	RF (SL)	99.6	99.6	99.6	99.6	–	–	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
11	Eddin et al. (2022)	Electricity Theft Attacks	Smart meter dataset from Ontario, Canada by Independent Electricity System Operator (IESO) (pri)	Bidirectional GRU (SL)	96.67	–	–	98.02	–	–	–	–
12	Esmalifalak et al. (2017)	Stealthy False Data Injection	Simulated on IEEE 118-bus system with MATPOWER (pri)	SVM (SL)	–	–	–	–	–	–	–	–
13	Zia et al. 2023)	False Data Injection Attacks (FDIAs)	Simulated on IEEE 14-bus system (pri)	Binary Relevance (SL)	95.1	95.5	85.5	91.6	–	–	–	–
14	Panthi (2020)	Power system, Cyber-Attack, Natural disturbances	Simulated on IEEE 3-bus system; Mississippi State University's Critical Infrastructure Protection Center (MSU-CIPC) power system dataset (pri)	RF (SL)	–	92.1 (Attack)	98.9 (Attack)	95.4 (Attack)	–	–	–	–
15	Sinioso-glou et al. (2021)	Operational anomalies, Cyber-Attack	Custom, SCADA (pub)	Auto-encoder GAN (UL)	94.7 (Avg.)	–	–	79.42 (Avg.)	-	-	81.2 (TP) 3.6 (FP)	-

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
16	Lu et al. (2019)	Non-technical losses (NTL): Fault or electricity theft	State Grid Corporation of China (SGCC) (pub)	DL Model (SSL)	95.1	94.4	80.4	86.6	0.964	–	–	–
17	Oprea and Bâra (2021)	Irregularities in electricity consumption, fraud in electricity consumption	Sourced from the Tunisian Company of Electricity and Gas (STEG) (pri)	Light GB (SL)	–	–	–	–	0.86	–	–	–
18	Wei et al. 2020)	Grid Load Anomaly Detection (GLAD)	Microgrid simulation from Mgsim (pub)	Time-series model (SL)	–	–	–	–	–	–	–	–
19	Hollingsworth et al. (2018)	Power anomaly detection	Sourced from Tennessee Valley Authority (TVA) (pri)	Autoregressive Integrated Moving Average + LSTM (SL)	92.1	–	–	–	–	93.9 (S)	69 (TP) 61 (FP)	–
20	Sun et al. (2022)	Smart meter data anomaly detection	State Grid Corporation of China (SGCC) (pub)	FCM (RL)	>88.2	–	–	–	>0.827	–	–	–
21	Jaiswal et al. (2021)	Smart meter data anomaly detection	Smart meter data from Ausgrid Corporation, Australia (pub)	RF (SL)	–	–	–	97.8	–	97.5 (S)	–	0.973 (G-Mean)

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
22	Cui et al. (2019)	Load Forecasting Under Cyber-Attacks	ISO New England (NEPOOL region) (pub)	k-Means (SL)	>93	–	–	–	–	–	–	–
23	Barua et al. (2020)	Spatial and temporal anomalies in current magnitude	µPMU dataset (req)	HTM (UL)	98	–	–	–	–	–	–	–
24	Ahmed et al. (2019)	Covert Data Integrity Assault	Historical SE-MF dataset simulated on IEEE 14, 39, 57, and 118-bus test systems (pri)	IF (UL)	94.5	–	–	94.1	–	–	–	–
25	Buzau et al. (2019)	Non-technical electricity losses	Sourced from Endesa Distribución, Spain (pri)	XGBoost (SL)	–	–	–	–	0.91	–	–	–
26	Hashatsi et al. (2020)	Non-technical losses—Electricity theft	Smart Meter Consumption Data collected from University of Massachusetts (req)	SVM (SL)	90.6	–	–	–		95.75 (DR)	–	–
27	Madhure et al. (2020)	Nontechnical losses, electricity theft	Sourced from the Commission of Energy Regulation (CER) and Irish Social Science Data Archive (ISSDA) (req)	CNN-LSTM (SL)	–	–	–	–	–	–	~5 (FP)	0.045 (MAE) 0.014 (MSE)

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S	TF	Other
										TP, TN	FP, FN	
28	Danilczyk et al. (2020)	Physical faults in the power system	Simulated from IEEE 9, and 39-bus benchmarks (pri)	CNN (SL)	95	–	–	–	–	–	–	–
29	Dai et al. (2020)	Power Consumption based	Collected power consumption side channel data (pri)	SVM (SL)	99.4	–	–	–	–	–	–	–
30	Wang et al. 2019a)	Electricity Theft	State Grid Corporation of China (SGCC) (pub)	LSTM (SL)	71	–	–	–	–	–	–	3.05 (RMSE)
31	Wang et al. (2019b)	Electricity Theft	Electricity Customer Behaviour Trial dataset (pub)	SVM-Autoencoder (RL)	–	–	–	–	–	93.4 (DR)	12.4 (FP)	–
32	Haq et al. (2023)	Electricity Theft	Smart meter data (req)	Deep CNN model (SL)	~93	~93	~96	~96	–	–	–	–
33	Javaid et al. (2021)	Electricity Theft	State Grid Corporation of China (SGCC) (pub)	CNN-LSTM-DSN (SL)	95.3	91.2	92.3	92.8	0.93	–	–	–
34	Zafar et al. (2023)	Electricity theft	Sourced from OEDI portal: theft detection dataset (pub)	ConvGRU (FL)	97.84	96.79	97	97.17	–	–	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
35	Zidi et al. (2023)	Electricity theft	Sourced from OEDI portal: theft detection dataset TDD2022 (pub)	RF (SL)	>85	–	–	>84	>0.92	–	–	–
36	Hasan et al. (2019)	Electricity theft	State Grid Corporation of China (SGCC) (pub)	CNN-LSTM (SL)	89	92	96	94	–	–	–	–
37	Karimipour et al. (2019)	Cyber-Attack	Simulated on IEEE 39, 118, and 2848-bus systems	Restricted Boltzman Machine (UL)	99	–	–	–	–	–	98 (TP) <2 (FP)	–
38	Kosek (2016)	Cyber-Attack	PV power production data (pub), Meteorological data (pub),	ANN (SL)	93 (Attack) ~99 (Normal)	–	–	–	–	–	–	–
39	Radoglou-Grammatikis and Sarigiannidis (2018)	Cyber-Attack	CIC-IDS2017 (pub)	DT (SL)	99.6	–	–	–	–	–	99.3 (TP)	-
40	Ünal et al. (2021)	Non-technical losses (NTL): False Data Injection Attacks, Cyber-Attack	Sourced from Turkish distribution network (pri)	DL (SL)	97.2	98.4	-	94.2	–	99.5 (S)	0.5 (FP)	97.2 (SEN)

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
41	Martinelli et al. (2022)	Intrusion Detection	Custom generated (pri)	AdaBoost (SL)	100	100	100	100	1	–	–	–
42	Khan et al. (2021)	Cyber-Attack: Intrusion Detection	KDD99 (pub), NSL-KDD (pub)	RF (SL)	~99	~99	~99	~99	–	~99 (DR)	–	–
43	Jithish et al. (2023)	Abnormal behaviors in the grid	KDD99 (pub), NSL-KDD (pub), CIDDS-001 (pub)	FedAVG (FL)	98.9	87.1	95	90.9	–	–	–	–
44	Moghaddass and Wang (2018)	Occurrence of anomalous events and abnormal conditions	Collected (pri)	Generative Model (SL)	93	97	–	–	–	98 (S) 72.47 (DR)	–	0.0919 (MAE) 0.87 (SEN)
45	Li et al. (2023)	Anomaly data in SGs	Collected from transformers (req)	Improved RF Algorithm (SL)	>95	–	–	>95	–	–	–	–
46	Yuan and Jia (2015)	Energy consumption	Sourced from Operation Energy Management System (OEMS) (pri)	Sparse Autoencoder (UL)	95.96	–	–	–	–	–	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S	TF	Other
										TP, TN	FP, FN	
47	Ford et al. (2014)	Energy fraud detection	Smart meter consumption data by the Irish Social Science Data Archive (ISSDA) (req)	ANN (SL)	-	–	–	–	–	–	25 (FP) 6.25 (FN) 93.75 (TP) 75 (TN)	–
48	Oprea et al. (2021)	Fraud detection	Sourced from the Commission of Energy Regulation (CER) and Irish Social Science Data Archive (ISSDA) (req)	Spectral Residual-CNN (SL)	90	87.5	–	89.4	–	–	–	–
49	Bashir et al. (2021)	Predict the stability of the SG	SG Dataset (pub)	Decision Tree (SL)	99.96	100	99.9	100	–	–	–	–
50	Kawoosa and Prashar (2022)	Non-Technical Losses (NTL)—Energy Theft	State Grid Corporation of China (SGCC) (pub), Smart meters in London (pub)	XGboost (SL)	94.73 (Mean)	95.07 (Mean)	94.7 (Mean)	93.47 (Mean)	–	93.15 (DR)	3.58 (FP)	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S	TF	Other
										TP, TN	FP, FN	
51	Sayin et al. (2023)	Concept Drift	MATPOWER simulation, sourced from the New York Independent System Operator (NYISO) (pub)	Decision Tree (SL)		99 (Avg.)	99 (Avg.)	99 (Avg.)	–	–	–	–
52	Vijayanand et al. (2019)	Intrusion detection system (IDS)—Cyber-Attack	CICIDS 2017 dataset (pub)	SVM (SL)	99.99	–	–	–	–	–	–	–
53	Mukherjee (2023)	Cyber-Attack: False Data Injection Attack (FDIA)	Simulated on IEEE 118-bus system with RTDS (pri)	CNN (SL)	~99	99.6	99.39	99.49	–	–	–	–
54	Nawaz et al. (2023)	Non-Technical Losses (NTL)—Electricity Theft	State Grid Corporation of China (SGCC) (pub)	CNN-XGB (SL)	92	92	54	15	–	–	–	–
55	Shabad et al. (2021)	Abnormal activities, Cyber-Attack	Sourced from Energy Systems Research Laboratory at Florida International University (pri)	IF (UL)	98.74	–	–	–	–	–	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S TP, TN	TF FP, FN	Other
56	Takiddin et al. (2022)	Electricity Theft, Cyber-Attack	State Grid Corporation of China (SGCC) (pub), Irish Smart Energy Trial (ISET) (req)	Deep Auto-encoders with LSTM (UL)	94.5	95	–	93.5	0.9	94 (DR)		5% (False Alarm)
57	Zhang et al. (2019)	Abnormal data of smart meters	Custom generated (pri)	Time-series model (SL)	-	–	–	–	–	–	–	–
58	Kurt et al. (2019)	Cyber-Attack	Simulated on IEEE 14-bus system with MATPOWER (pri)	Partially Observable Markov Decision Process (RL)		~99	~1	~99	–	–	–	
59	Mukherjee et al. (2022)	Cyber-Attack: False Data Injection Attack (FDIA)	Custom (req)	CNN (SL)	>95	–	–	–	0.54	–	–	0.000057 (RMSE) $0.3249 \times 10{-}8$ (MSE) 0.00872 (MAE)
60	Jindal et al. (2016)	Non-Technical losses—Electricity theft	Commercial and Residential Hourly Load Profiles for all TMY3 Locations in the United States (pub)	SVM (SL)	92.5	–	–	–	–	–	5.12 (FP)	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S	TF	Other
										TP, TN	FP, FN	
61	Menon and Radhika (2016)	SG traffic and Cyber-Attack	Raw SG data (pri)	k-Means (UL)	–	–	–	–	–	–	–	Found 0.91% anomalies in 5 days data
62	Zhang et al. (2019)	Cyber-Attack	Custom simulated (pri)	PSO-SVM (SL)	–	–	–	–	–	–	–	–
63	Alani et al. (2023)	Cyber-Attack	DNP3 intrusion detection dataset (pub)	RF (SL)	99.5 (Detection)	–	–	99.5 (Detection)	–	–	–	–
				XGboost (SL)	98.82 (Classification)			98.82 (Classification)				
64	Han et al. (2023)	False Data Injection Attack (FDIA)	Simulated on IEEE 14-bus, 118-bus, and 300-bus power system with MATPOWER (pri)	Modified Temporal Multi-graph Convolutional Network (SL)	98.3 (14-bus)	99.8 (14-bus)	86.4 (14-bus)	92.6 (14-bus)	–	–	–	–
					96.4 (118-bus)	98.9 (118-bus)	72.0 (118-bus)	83.3 (118-bus)				
					95.8 (300-bus)	96.7 (300-bus)	68.5 (300-bus)	80.2 (300-bus)				
65	Mohammadpourfard et al. (2021)	Cyber-Attack	New York independent system operator (NYISO, pub); Simulated on IEEE 14-bus system with MATPOWER (pri)	LSTM-RNN (SL)	–	–	–	>90	–	–	<0.4 (FP)	–
66	Wang et al. (2019)	Power grid disturbances and Cyber-Attack	ICS dataset (pub)	AdaBoost (SL)	93.91	93.8	93.6	93.5	—	93.6 (DR)	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S / TP, TN	TF / FP, FN	Other
67	Mokhtari et al. (2021)	Abnormal activities, Intrusion Detection, Cyber-Attack, fault detection, event detection	HAI (HIL-based Augmented ICS) Security Dataset (pub)	RF (SL)	99.76	99.76	99.76	99.76	0.9976	–	–	–
68	Ida Evangeline et al. (2024)	Electric power anomaly detection	State Grid Corporation of China (SGCC) (pub)	Trace-based graph DL mode (SL)	–	99.2	98.4	94.6	–	–	–	–
69	Akagic and Džafić (2024)	Anomaly Detection and Identification Module	Simulated on IEEE 6-bus, 14-bus, and 30-bus system (pri)	DL model (SL)	90.05	99.9	99.9	99.9	–	–	–	–
70	Alshehri et al. (2024)	Cyber Attacks	Sourced from the Commission of Energy Regulation (CER) and Irish Social Science Data Archive (ISSDA) (req)	FL	84.17	80.01	85.11	82.48	0.873		–	–
71	Sharma and Tiwari (2024)	Electricity Theft	Irish Smart Energy Trial (ISET) (req)	Optimized XGBoost (SL)	92.42	92.19	92.42	92.24				

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S / TP, TN	TF / FP, FN	Other
72	Priyadarsini and Sonekar (2025)	Cyber attacks	Inferred use of device data matrices: sensors, generators, substations, breakers (pri)	Custom CNN (SL)	93-95	–	–	–	–	–	–	–
73	Ramul et al. (2025)	Cyber attacks	Simulated on IEEE 69-bussystem (pri)	Hybrid Fuzzy Multi-Agent Deep RL (RL)	>99	–	–	–	–	–	–	–
74	Elshennawy et al. (2025)	Electricity thefts and energy fraud	State Grid Corporation of China (SGCC) (pub)	CNN+LSTM (SL)	98.75	–	95.45	97.7	–	–	–	–
75	Wang et al. (2025)	Cyber attacks	Simulated on IEEE 14-bus & IEEE 118-bus system (pri)	Spatial feature-based temporal convolutional network (SL)	99.68 (14-bus), 97.20 (118-bus)	99.90 (14-bus), 96.80 (118-bus)	98.83 (14-bus), 91.85 (118-bus)	99.36 (14-bus), 94.26 (118-bus)	–	–	–	–

Table 7 (continued)

No.	References	Anomaly type	Dataset used	Model	Proposed/Best Model Results Reported							
					Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	DR, S	TF	Other
										TP, TN	FP, FN	
76	Sinha et al. (2021a)	Cyber attacks	Simulated on IEEE 9-bus & IEEE 14-bus system	Cyber Physical Defense Framework (One-Class SVM) SL+UL	Up to 96.9 (abnormal), 96.7 (normal)	–	–	–	–	–	FP: 6.62 (9-bus), 7.03 (14-bus) FN: 6.54 (9-bus), 6.75 (14-bus)	–
77	Sinha et al. 2021b)	Cyber attacks	Simulated smart grid data on:3-bus, IEEE 9-bus, 14-bus, and 30-bus systems	Deep RL (RL)	–	–	–	–	–	–	–	Multiple RL agents trained per grid
78	Sinha et al. (2022)	Cyber attacks & Electricity thefts and energy fraud	ICS dataset (pub), State Grid Corporation of China (SGCC) (pub)	IDS: Lightweight model (not named) Theft: 1D-CNN, ANN, and classifiers (CatBoost best) (SL)	99.4	–	–	–	–	–	–	–

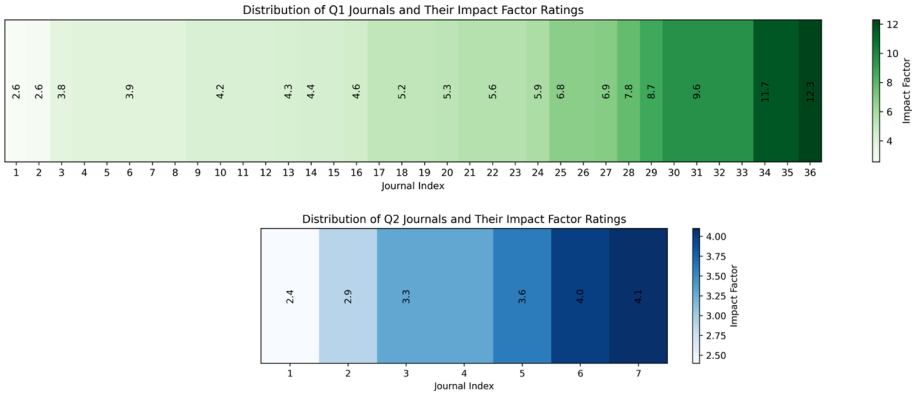


Fig. 3 Distribution of Q1 and Q2 journals and their impact factor ratings

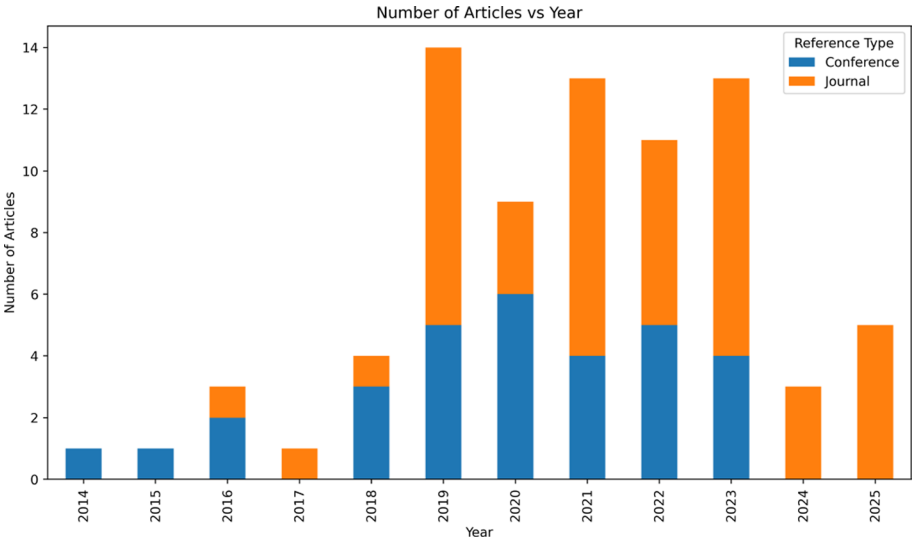


Fig. 4 Number of articles vs year

5.1 Overview

This section overviews all the papers selected through our thorough research methodology. Figure 4 illustrates the rising number of articles published yearly, with a rise in 2019. Among the publications, 32 are conference papers, and 46 are journal articles. Figure 5 illustrates the citation numbers for each of the selected papers.

5.2 Analysis of gathered data

Researchers proposed several ML algorithm-based solutions to identify a wide variety of anomalies in SGs or smart meters. Cyber-attack detection seems to be a widely discussed

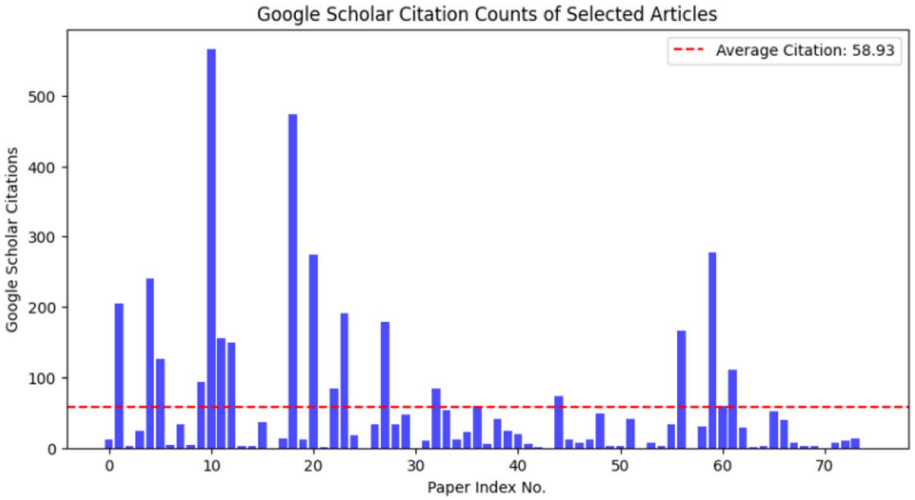


Fig. 5 Google scholar citations of the papers

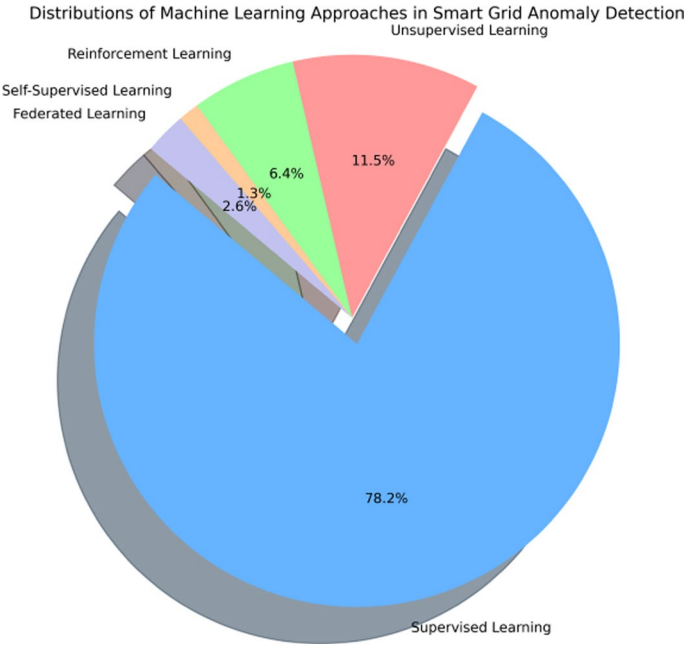


Fig. 6 Distribution of ML model types used in the articles

topic when it comes to identifying anomalies in SGs. Additionally, these papers cover methods for detecting other anomalies, including electricity theft, anomalous data patterns, energy fraud, spatial and temporal irregularities, physical faults, concept drift, and predictive maintenance. Figure 6 depicts the distribution of ML model types used in the articles. In our review, 61 out of 78 papers investigating anomaly detection in SGs using ML algo-

rithms favored SL, while 9 favored UL. Five papers employed RL, 1employed SSL, and 2 employed Federated Learning (FL).

Figure 7 utilized the distribution of the ML algorithm used in the papers. It is visible from Fig. 7 that SVMs and CNNs were the most popular choices among the options, both appearing in 23 papers. On the other hand, LSTMs accounted for 16 occurrences. In addition, RF and DT were frequently employed, appearing in 17 papers each. Ten papers used GNB and XGBoost. There were 14 papers utilizing KNN and six papers using RNN. PCA and IF were each presented in five papers. There are three papers that use K-Means Clustering, while 23 papers used or proposed algorithms that belong to the category labeled "Other," indicating a variety of algorithms other than those specified.

In Table 7, 78 ML-based solutions that are designed to detect anomalies in SGs and smart meters are presented. Several details are provided in the analysis, including the type of anomaly, the dataset used for training and testing, ML model types, primary algorithms used to detect anomalies, and results (proposed model evaluation metrics). Diverse solutions that illustrate popular algorithms that can be used to detect anomalies in SGs and smart meters are presented here. Table 7 reports the proposed solutions that address anomaly detection in SGs and the techniques used. To evaluate the performance of their algorithms, researchers in the field of anomaly detection for SGs utilize a variety of metrics. Accuracy, which measures overall correctness, and Precision, which measures the accuracy of positive predictions, are a few of these metrics. An AUC indicates the model's ability to distinguish between classes by measuring its ability to capture all actual positives, whereas the F1-Score balances Precision and Recall. A number of metrics have been used to evaluate the detection of electricity theft, including Theft Confidence (TF), Detection Rate (DR), Specificity (S), False Positives (FP), False Negatives (FN), True Positives (TP), and True Negatives (TN). The Detection Rate measures the algorithm's accuracy in identifying anomalies, while the specificity measures its accuracy in identifying non-anomalous instances. To evaluate prediction accuracy, metrics including MSE, MAE, and RMSE are used.

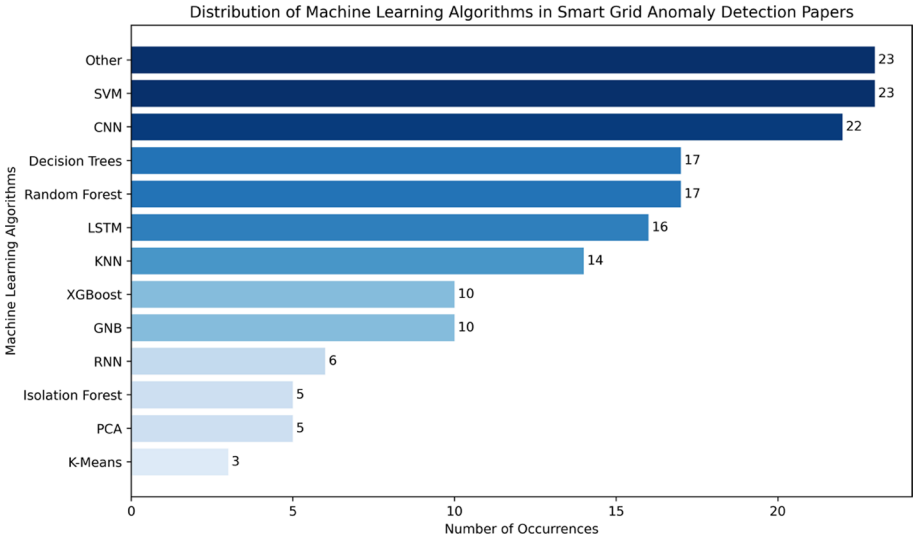


Fig. 7 Distribution of ML algorithms used in the papers

5.3 Artificial intelligence models to detect anomalies in smart grid

A variety of solutions have been proposed for detecting various anomalies in power generation and distribution in SG research, which is a dynamic and growing field. The studies investigated many anomalies, including cyber-attacks, electricity theft, data anomalies, physical faults, and predictive maintenance.

To address these challenges, ML is crucial, with methods such as neural networks, RF, and RL providing improved reliability and security. Various solutions are provided to detect the multiple anomalies listed in Table 7 in the research articles reviewed in this review. A comprehensive overview of various types of anomalies in SG systems is provided in Fig. 8, which systematically categorizes these anomalies.

Figure 9 gives a comprehensive overview of the distribution of anomaly events in the articles.

5.3.1 Cyber attacks

Most studies focus on SG cyberattacks, which account for around 47% of all anomalies. Several ML techniques are employed in these studies to detect and mitigate cyber threats,

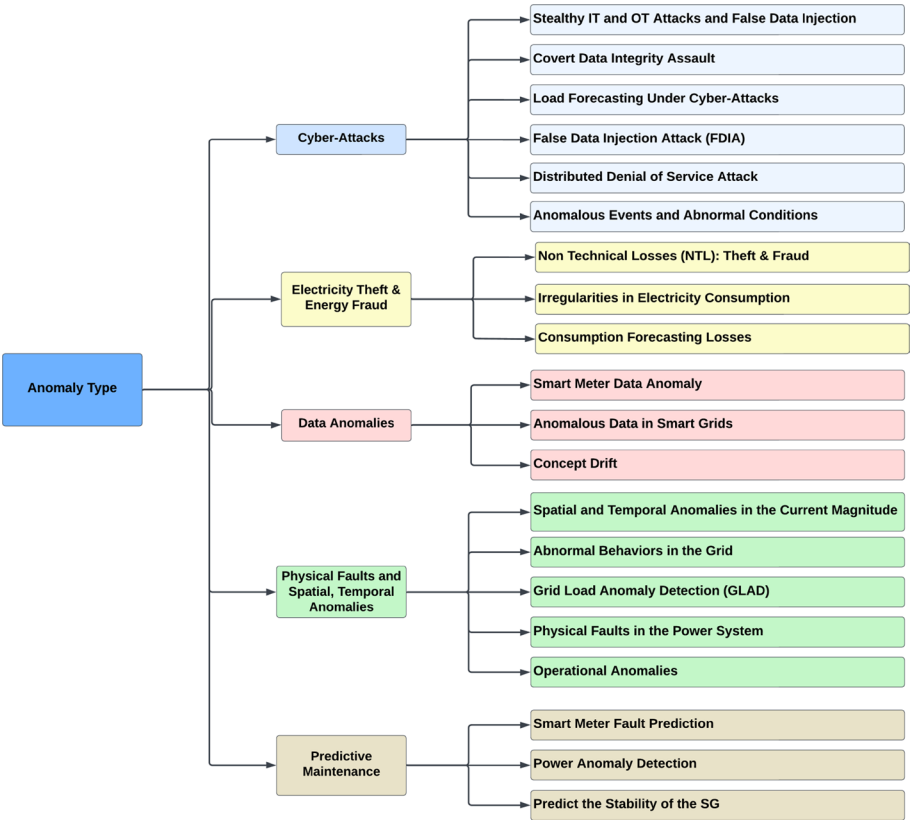


Fig. 8 Anomaly types

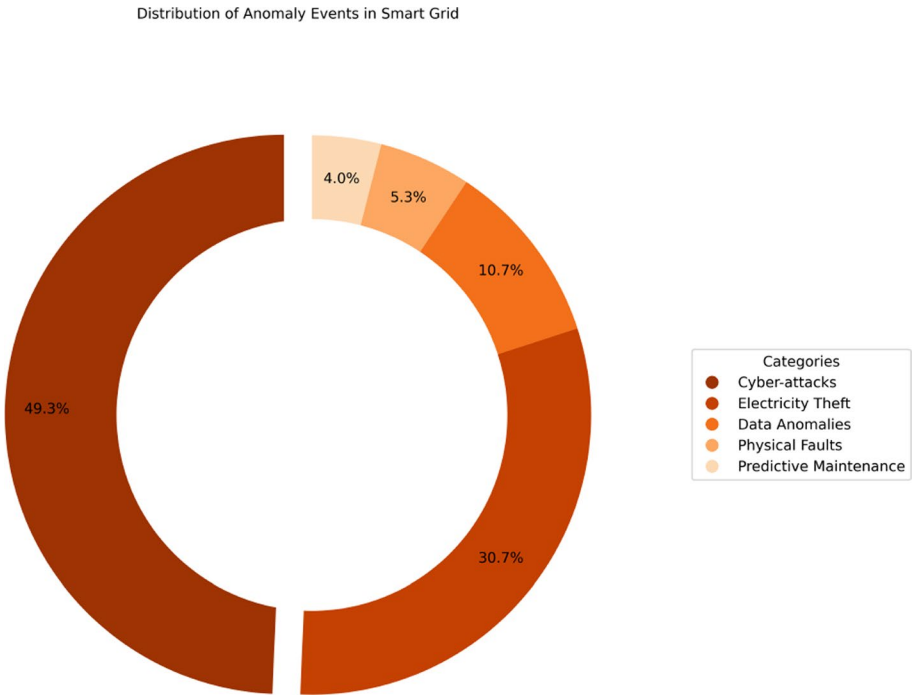


Fig. 9 Distribution of anomaly events in the articles

emphasizing the importance of securing SG communication networks. The comprehensive literature analysis of Table 7 on SG cyberattacks is given below:

- Techniques: Several ML algorithms, such as RF, LSTM, DT, IF, SVM, kNN, NB, MLP, RNN, Auto-encoders, DBNs, Symbolic Dynamic Filtering, and RL, are utilized (Elmrabit et al. 2020; Ali et al. 2022; Hyder et al. 2023; Diaba and Elmusrati 2023; Eddin et al. 2022; Cui et al. 2019; Ahmed et al. 2019; Karimipour et al. 2019; Kosek et al. 2016; Mukherjee 2023; Takiddin et al. 2022; Mukherjee et al. 2022; Menon and Radhika 2016; Han et al. 2023; Alshehri et al. 2024).
- Detection Methods: The methods encompass anomaly detection, theft detection, CDIA detection, load forecasting, and energy fraud detection.
- Performance: The performance is characterized by high detection rates, with techniques like RNN-based detection (Eddin et al. 2022), deep autoencoders with LSTM (Takiddin et al. 2022), and RF (Elmrabit et al. 2020) displaying accuracy levels ranging from 95% to 99.76%.
- Applications: These techniques find practical applications in areas such as AMIs (Takiddin et al. 2022), DERs (Ahmed et al. 2019), SCADA systems (Mokhtari et al. 2021), and large-scale SGs.
- Challenges: The main challenges faced include difficulties in detecting stealthy and covert attacks and distinguishing between cyber-attacks and natural disturbances.

5.3.2 Electricity theft and energy fraud

Considerable attention has been devoted to the issue of combating electricity theft and energy fraud, as evidenced by the allocation of 22 out of the 78 papers to this particular topic. One proposed solution aims to leverage a range of machine-learning algorithms to identify abnormal consumption patterns within SGs, thereby bolstering their security. A comprehensive analysis of the literature, as presented in Table 7, focuses specifically on the problem of SG electricity theft and energy fraud.

- Techniques: ML algorithms employed in this context include LSTM, CNNs, SVMs, XGBoost, K-means, Synthetic Minority Oversampling Technique (SMOTE), PCA, FL, and a combination of CNN and LSTM (Fenza et al. 2019; Wang et al. 2019; Sharma and Tiwari 2024; Haq et al. 2023; Zafar et al. 2023; Zidi et al. 2023; Hasan et al. 2019) (Radoglou-Grammatikis and Sarigiannidis 2018; Kawoosa and Prashar 2022; Priyadarsini and Sonekar 2025).
- Detection Methods: The methods employed for identifying electricity theft encompass profiling electricity consumption, anomaly detection, K-SMOTE-PCA-XGBoost for energy theft, and Convolutional Gated Recurrent Units (ConvGRU) for FL (Zafar et al. 2023; Kawoosa and Prashar 2022).
- Performance: The models, such as LSTM, CNN, and ConvGRU, exhibit high accuracy rates ranging from 78% to 97.84%, surpassing the performance of traditional methods.
- Applications: The detection of electricity theft primarily relies on analyzing data from smart meters and identifying anomalies in energy consumption patterns.
- Challenges: Key challenges in this area include addressing class imbalances in the available data, ensuring data privacy, and mitigating high false-positive rates observed in certain models.

5.3.3 Data anomalies

Researchers utilize smart meters and grid data to enhance the accuracy and efficiency of anomaly detection. Several studies briefly discuss the anomalies in the data.

- Techniques: The techniques employed include ML approaches such as hierarchical distributed fog computing, XGBoost, DT, GMM-LDA clustering, PSO-SVM, FL, 1D-CNN, and Deep RL (Jaiswal et al. 2020, 2021; Sun et al. 2022; Buzau et al. 2019; Jithish et al. 2023; Li et al. 2023; Sayin et al. 2023; Zhang et al. 2019; Ida Evangeline et al. 2024; Akagic and Džafić 2024).
- Detection Methods: The methods used for real-time anomaly detection in smart meters involve supervised and unsupervised learning models and the implementation of fog computing architecture (Jaiswal et al. 2020) (Buzau et al. 2019).
- Performance: The performance evaluation indicates high G-mean scores (97.3%) and excellent detection accuracy for models such as IRFA, PSO-SVM, and FL-based methods (Jaiswal et al. 2021).
- Applications: The applications of anomaly detection in smart meter data include identifying abnormal consumption patterns and detecting non-technical losses.
- Challenges: The challenges in this field include balancing measurement granularity and

privacy concerns and adapting to dynamic network changes.

5.3.4 Physical faults and spatial temporal anomalies

Four out of 67 studies specifically apply to SG's physical faults and technical losses. In these studies, neurocognitive architectures, DL algorithms, and hierarchical frameworks are introduced to identify anomalies in SGs that are effective. A comparative analysis is given below.

- Techniques: Neuro-cognitive architectures, CNNs, hierarchical frameworks, DL algorithms, the GLAD method, and the extreme studentized deviation test (Barua et al. 2020; Danilczyk et al. 2020; Dai et al. 2020; Moghaddass and Wang 2018).
- Detection Methods: Real-time, unsupervised anomaly detection utilizing PMU data, DL for power systems, hierarchical frameworks for smart metering data.
- Performance: Nearly perfect fault detection accuracy, 99.4% detection accuracy in hierarchical frameworks, and high true positive rates with low false positive rates using GLAD.
- Applications: Fault detection in power systems, anomalies in relay protection, subsystem and system-level anomaly detection.
- Challenges: Handling high dimensionality, noise robustness, and effective diagnosis with multiple measurements.

5.3.5 Predictive maintenance

Three studies explore predictive maintenance, using ML algorithms for predicting fault types, forecasting losses, and ensuring SG stability.

- Techniques: The techniques employed in this study include DT, NB, SVM, Logistic Regression, ARIMA, LSTM, and combined method models (Helong et al. 2019; Matijašević et al. 2022; Hollingsworth et al. 2018; Bashir et al. 2021).
- Detection Methods: The detection methods used in this research encompass predicting SG stability, forecasting faults in energy meters, and anomaly detection in energy consumption data.
- Performance: It has been observed that DTs exhibit superior accuracy, precision, recall, and F1-scores. Furthermore, combining ARIMA and LSTM models demonstrates robust anomaly detection capabilities.
- Applications: The applications of this study mainly revolve around fault type prediction, energy consumption analysis, and low-voltage distribution network loss prediction.
- Challenges: The challenges encountered in these studies involve analyzing real failure data, overcoming obstacles associated with network data management, and ensuring the reliability and robustness of the prediction models.

Figure 10 visualizes the ML model types used in the reviewed papers for different types of anomalies. The proposed methods show promise in enhancing security, detecting anomalies, and ensuring the reliability of SG systems. These studies demonstrate that ML can be

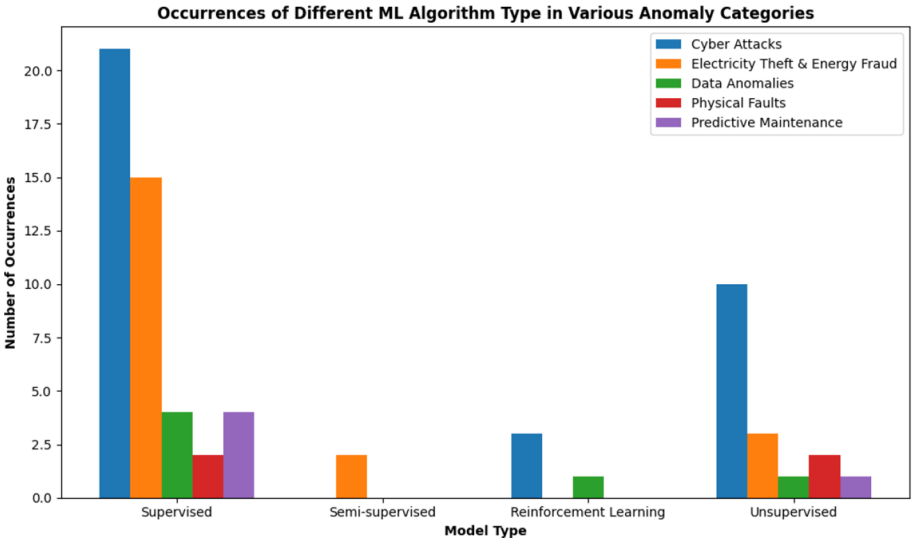


Fig. 10 ML model types are used in the research for different types of anomalies

used to address a wide range of challenges associated with SGs as a whole. Research and innovation in this field are essential to ensuring the future of smart energy networks.

6 Discussion

The research areas are explored here based on the analyzed data.

6.1 Trends in AI for smart grid anomaly detection

Increasingly, researchers are exploring innovative approaches for enhancing grid security and reliability through AI techniques for anomaly detection in SGs. The preference for techniques that do not require labeled training data is one notable trend that indicates a shift toward UL approaches. As an example, iForests (Ahmed et al. 2019), symbolic dynamic filtering using DBNs (Karimipour et al. 2019), and clustering using k-means (Menon and Radhika 2016) are gaining popularity due to their ability to detect anomalies without the need for labeled data. In order to achieve high accuracy and F1-scores, iForest requires fewer training samples, making it more effective for anomaly detection in large datasets. However, it may suffer from poor performance when the data is highly imbalanced, and it may not be as effective when detecting subtle anomalies without adequate tuning (Ahmed et al. 2019). Recent studies have shown significant improvements in classification accuracy with a hybrid approach combining CNNs and LSTM networks (Hasan et al. 2019). Using time-series data, these models can identify irregular energy consumption patterns, signaling a shift towards more nuanced and context-aware anomaly detection algorithms. Despite their ability to capture both spatial and temporal patterns in data, these models are difficult to train and require extensive computational resources, making them less suitable for real-time applications in resource-constrained environments. Research has demonstrated the

effectiveness of data augmentation and sophisticated feature extraction methods in improving the performance of ML models (Abdelkhalek et al. 2022).

In order to identify the most effective ML algorithms, researchers conduct comprehensive evaluations and comparisons. (Elmrabit et al. 2020; Hyder et al. 2023) compare algorithms such as SVM, kNN, NB, MLP, and RF. To capture complex patterns and temporal dependencies in SG data, advanced neural network architectures such as DBNs, Autoencoder-GAN, and LSTM-RNN are being integrated.

Novel approaches are being developed to improve anomaly detection in areas such as fraud, theft, and concept drift. To improve accuracy, sophisticated ML models, such as LSTM networks (Fenza et al. 2019; Wang et al. 2019), and ensemble models (Madhure et al. 2020; Javaid et al. 2021), are employed. However, this integration of advanced ML techniques entails challenges related to model complexity, interpretability, and the requirement for extensive computational resources. Several techniques are being used to address class imbalances, including the Synthetic Minority Oversampling Technique (SMOTE) (Hasan et al. 2019) and Adaptive Synthesis (ADASYN) (Javaid et al. 2021), highlighting the importance of providing balanced datasets for model training. In addition, privacy-preserving techniques are being incorporated, as demonstrated in the proposed deep FL mechanism (Mukherjee et al. 2022) that protect user information while enabling collaborative learning across distributed data sources. FL ensures that sensitive consumer information is not exposed during the learning process by keeping data localized and only sharing model updates.

To demonstrate ML's ability to adapt to dynamic network changes, adaptive methods such as GMM-LDA clustering feature learning and PSO-SVM (Zhang et al. 2019) are presented to detect abnormal data in smart meters. As consumer behavior and consumption patterns change over time, resulting in concept drift that adversely affects static models, adaptive learning methods like GMM-LDA clustering feature learning combined with PSO-SVM adjust dynamically to new patterns in the data, maintaining model accuracy.

Neurocognitive architectures (Baru et al. 2020), DL approaches (Danilczyk et al. 2020), hierarchical frameworks (Moghaddass and Wang 2018), and methods tailored to specific aspects of anomaly detection are characteristics of ML techniques for SGs. A major objective of this shift is to improve the detection of faults and cyber-attacks within SGs by capturing spatial and temporal data patterns more effectively (Danilczyk et al. 2020; Wang et al. 2019).

As SG environments become increasingly complex and interconnected, threats continue to evolve. Future research should focus on developing adaptive, resilient models capable of detecting various anomalies, including zero-day attacks and sophisticated cyber threats (Abdelkhalek et al. 2022; Ali et al. 2022). However, the complexity arises from the diverse nature of datasets encountered in SG environments, making it challenging to suggest a single model for a specific task. Recognizing these challenges, we present Table 8 as a guideline rather than a universal solution to highlight the importance of adapting ML approaches to the unique characteristics of individual datasets.

6.2 Types of anomalies

A wide range of anomalies are identified in the SG anomaly detection studies, providing comprehensive solutions for enhanced security and reliability. Cyber-attacks are a signifi-

Table 8 Summary of ML techniques for anomaly detection in SGs

Study references	ML technique used	Dataset Used	Key Findings	Accuracy	False Positive Rate (FPR)	Remarks
Banik et al. (2023), Al-Tamimi et al. (2024), Jaiswal et al. (2021)	iForest, SVM, kNN, NB, MLP	KDD-CUP 99, NSL-KDD	iForest-based scheme outperforms other ML techniques	High	Low	iForest requires fewer samples for higher accuracy
Hollingsworth et al. (2018a, Sun et al. (2022), Yang et al. (2011)	HTM, XGBoost, NB	µPMU data, On-field inspection data	HTM architecture achieves high accuracy for real-time detection	96–98%	Very Low	HTM offers competitive real-time anomaly detection
Wang and Lu (2013), Barua et al. (2020), Ahmed et al. (2019)	SVM, CNN, RNN	PMU voltage data, Solar and wind energy profiles	DL models like CNN show near-perfect fault detection accuracy	Near 100%	–	DL techniques show promising results
Abdelkhalek et al. (2022), Sinha et al. (2024), Marugán et al. (2018)	RF, SVMs, LSTM	UNSW-NB15, IEEE standard test systems	RF algorithm provides the best performance among ML algorithms	High	Low	RF is recommended for its overall performance
Jaiswal et al. (2020), Cui et al. (2019), Radoglou-Grammatikis and Sarigiannidis (2018)	Neural Networks, Cubic SVM, NB	Energy consumption data	Neural networks achieve high detection rate for energy fraud	Up to 90%	–	Neural networks are effective in detecting fraudulent activities
Elmrabit et al. (2020), Diaba and Elmusrati (2023), Siniosoglou et al. (2021)	ARIMA, LSTM, Decision Tree	Energy consumption data, Co-simulation SG platform	LSTM and DT show robust performance in anomaly detection	High	–	Emphasis on the effectiveness of time series analysis

cant focus, with approximately 47% of the studies covering this type of anomaly highlighting the prevalence and diversity of cyber threats. Figure 11 depicts occurrences of each type of anomaly detected in selected articles. The proposed systems often employ ML algorithms to detect stealthy and sophisticated cyber-attacks, such as false data injection, man-in-the-middle attacks, and denial-of-service attacks. For example, (Ahmed et al. 2019) and (Karimipour et al. 2019) discuss how the advancements in UL can be used to identify covert attacks on data integrity without the need for labeled data. Furthermore, (Mohammadpourfard et al. 2021) introduces a DL-based model incorporating LSTM networks to distinguish cyberattacks from natural grid fluctuations.

For detecting abnormal consumption patterns and ensuring the security of SGs, several ML algorithms have been proposed, such as LSTM networks, CNNs, SVMs, and FL methods (Fenza et al. 2019; Lu et al. 2019; Wang et al. 2019; Yuan and Jia 2015; Kawoosa and Prashar 2022; Haq et al. 2023; Zafar et al. 2023; Zidi et al. 2023; Hasan et al. 2019). Accordingly, (Buzau et al. 2019) discusses the implementation and application of XGBoost

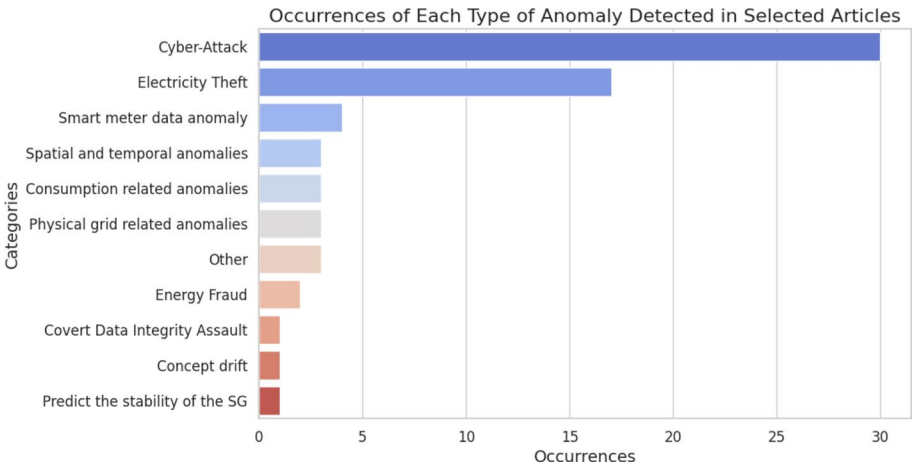


Fig. 11 Occurrences of each type of anomaly detected in selected articles

classifiers in real NTL campaigns. By incorporating LSTM models, as shown in Wang et al. 2019, abnormal consumption patterns can be detected more accurately than with conventional ARIMA models. Moreover, (Haq et al. 2023) and (Hasan et al. 2019) demonstrate how DL models, such as CNN-LSTM, can be used to detect irregular consumption patterns, indicating a shift toward more sophisticated, data-driven approaches for detecting thefts. Data anomalies are addressed through innovative ML-based approaches, such as hierarchical distributed fog computing, real-time anomaly detection, forecasting, and labeling power data with anomalies (Jaiswal et al. 2020; Sayin et al. 2023; Jaiswal et al. 2021; Jithish et al. 2023; Li et al. 2023).

Various neuro-cognitive architectures, DL algorithms, and hierarchical frameworks are employed to detect physical faults and spatial–temporal anomalies in SGs (Wei et al. 2020; Barua et al. 2020; Danilczyk et al. 2020; Dai et al. 2020; Moghaddass and Wang 2018). As an illustration of how statistical methods may be applied to improve grid security, (Wei et al. 2020d) introduces a novel GLAD method for detecting microgrid anomalies. Furthermore, DL-based anomaly detection algorithms discussed in Danilczyk et al. 2020 are able to detect anomalies using PMU voltage data, providing high accuracy in fault location.

The key literature highlights several strengths and weaknesses of the methods employed. For instance, the use of UL techniques such as IFs (Ahmed et al. 2019) and symbolic dynamic filtering using DBNs (Karimipour et al. 2019) are notable for their ability to detect anomalies without labeled data. However, these methods may struggle with identifying complex patterns without sufficient contextual information. Similarly, hybrid models like CNN-LSTM (Hasan et al. 2019) offer high accuracy by capturing spatial and temporal data but have increased computational demands and complexity. FL (Zafar et al. 2023) (Jithish et al. 2023) effectively addresses privacy and confidentiality issues by enabling collaborative learning across distributed data sources without sharing raw data. However, coordinating multiple distributed nodes and the overhead associated with FL pose significant challenges. The integration of advanced neural network architectures like LSTM (Ida Evangeline et al. 2024) and CNN (Haq et al. 2023) demonstrates the efficacy of DL models in detecting sophisticated anomalies.

These models excel in capturing temporal dependencies and complex patterns in the data. Nevertheless, their complexity and the requirement for extensive computational resources can limit their practical applicability. The challenges associated with using time series data of consumers in the energy market include handling the high dimensionality of the data, ensuring real-time processing capabilities, and ensuring data privacy. However, ensuring the scalability and efficiency of these models in real-world scenarios remains an ongoing research focus.

6.3 AI models for detecting smart grid anomalies

SG anomaly detection utilizes a variety of ML models to address different types of anomalies. Various algorithms have been employed for cyber-attacks, which constitute a substantial portion of the studies. Unsupervised ML techniques, such as iForest (Ahmed et al. 2019), have been compared with supervised methods, such as SVMs, kNN, NB, and MLP (Ahmed et al. 2019). Other supervised classification models examined include RF (Elmrabit et al. 2020), Decision Tree (Hyder et al. 2023), Ada Boost (Hyder et al. 2023), and Logistic Regression (Hyder et al. 2023). In detecting the DDoS attacks on SG communication infrastructure, ensemble learning approaches (Ashrafuzzaman et al. 2020) and hybrid DL algorithms (Diaba and Elmusrati 2023), which combine CNNs with Gated Recurrent Units (GRUs), have demonstrated superior performance.

Using ML regression models, a hierarchical distributed Fog Computing architecture incorporating ML regression models (Jaiswal et al. 2020) is demonstrated to detect anomalies in Smart Meter data in real-time. To distinguish honest customers from those with abnormal meters, XGBoost classifiers were employed in another study (Buzau et al. 2019). Various approaches for detecting anomalies in SG data include decision tree algorithms (Sayin et al. 2023) and UL models using GMM-LDA clustering feature learning and PSO-SVM (Zhang et al. 2019).

The problem of physical faults and spatial and temporal anomalies is addressed through neuro-cognitive architectures like HTM (Barua et al. 2020) and DL algorithms like CNNs (Danilczyk et al. 2020). By utilizing data from PMUs, these models are exceptionally effective for unsupervised, real-time anomaly detection. Using the hierarchical framework described in Moghaddass and Wang (2018), effective anomaly detection at the subsystem and system levels is demonstrated by addressing the challenges of high dimensionality and missing data. ML algorithms such as ARIMA and LSTM can be used to predict fault type, as well as combined methods to improve accuracy (Hollingsworth et al. 2018). Using ML-based forecasting techniques for losses in a low-voltage distribution network (Matijašević et al. 2022) demonstrates the applicability of these techniques to ensuring that SGs are reliable and stable.

A critical analysis of ML methods for SG anomaly detection reveals several strengths and weaknesses. Unsupervised learning techniques, such as IFs (Ahmed et al. 2019) and DBNs (Karimipour et al. 2019), are advantageous for detecting anomalies without labeled data, but they may lack context specificity. SL models like RF (Elmrabit et a. 2020), DT (Hyder et al 2023), and SVMs (Ahmed et al. 2019) offer high accuracy but require extensive labeled datasets and can be computationally demanding. DL architectures, including LSTM networks (Ida Evangeline et al. 2024), CNNs (Haq et al. 2023), and hybrid models (Diaba and Elmusrati 2023), excel in capturing complex patterns but come with increased compu-

tational costs and implementation complexity. FL techniques, such as deep FL (Zafar et al. 2023), address privacy concerns but pose challenges in coordination and overhead management. Real-time detection methods, including hierarchical and fog computing architectures (Jaiswal et al. 2020; Wei et al. 2020), provide real-time capabilities but may struggle with scalability and data integration. Models like LSTM and CNN-LSTM effectively capture temporal dependencies but require significant computational resources. It is essential to recognize the variety of datasets observed while also considering the relationship between different forms of anomalies and the ML techniques used to detect them. Table 9 provides detailed information on the type of anomaly detected, ML techniques utilized, datasets employed (see Table 7 for datasets' sources), and detection accuracy.

6.4 Result analysis of anomaly detection

The results of the anomaly detection algorithms presented in the reviewed papers indicate overall satisfaction in terms of accuracy and effectiveness. As an example, the proposed iForest-based scheme (Ahmed et al. 2019) demonstrated superior performance to traditional ML algorithms such as SVM, kNN, NB, and MLP, achieving better accuracy and F1-scores while requiring fewer training samples for the detection of covert data integrity attacks in SG communication networks. Similarly, the RF algorithm emerges as a standout performer in cyber-attack detection (Elmrabit et al. 2020), demonstrating its superiority compared to the other methods evaluated regarding accuracy, precision, recall, and AUC. Additionally, the ML-based anomaly detection system presented in Hyder et al. 2023 for critical cyber-physical infrastructure demonstrates robust performance with high accuracy, precision, recall, and F1-scores. Several papers have proposed unsupervised machine-learning approaches to detect anomalies, cyber-attacks, and disruptions in SGs (Karimipour et al. 2019; Menon and Radhika 2016) (Shabad et al. 2021). These approaches demonstrate strong performance in detecting anomalies, cyber-attacks, and disturbances in SGs.

For example, (Fenza et al. 2019) achieves a precision of 78% and a recall of 88% in identifying anomalies in SG data in the area of electricity theft and energy fraud detection. As reported in Zhang et al. 2018, the semi-supervised generative Gaussian mixture model can detect energy theft with superior accuracy and minimize false alarms. It has been demonstrated that approaches using DL models such as CNN, LSTM networks, and hybrid models combining multiple ML techniques can effectively detect complex energy consumption patterns indicative of fraud or system malfunctions. These models, tested on large datasets, have shown high degrees of accuracy, with deep FL models achieving up to 97% accuracy in specific cases (Zafar et al. 2023), as have other approaches utilizing ML and DL that have demonstrated similar performance levels (Abdelkhalek et al. 2022, Hyder et al. 2023; Ahmed et al. 2019).

Anomalies detection algorithms for smart meter data show promising results as well. For example, (Jaiswal et al. 2020) combines a hierarchical distributed Fog Computing architecture with ML-based anomaly detection, achieving superior accuracy, precision, recall, and F1-scores over existing methods. To detect anomalies and contingencies within the SG, the authors in Sayin et al. 2023 utilize supervised decision tree algorithms. The proposed approach effectively handles unexpected situations by demonstrating the stability of accuracy under normal operating conditions and its ability to identify anomalies when accuracy deviates. Despite these successes, there remain limitations, particularly in terms of general-

Table 9 Overview of anomaly detection types and approaches in SG systems

Study references	Types of anomaly detected	ML techniques used	Datasets used	Detection accuracy
Banik et al. (2023), Al-Tamimi et al. (2024), Sun et al. (2022), Jaiswal et al. (2021), Yang et al. (2011)	Electricity Theft	CNN, LSTM, RF, XGBoost, etc	Real-world and synthetic datasets	Varies; up to 98.4%
Abdelkhalek et al. (2022), Sinha et al. (2024), Marugán et al. (2018), Cui et al. (2019), Radoglou-Grammatikis and Sarigiannidis (2018)	Cyber Attacks (DoS, FDI, Stealthy Attacks, etc.)	SVM, kNN, RF, Deep Neural Networks, etc	KDDCUP 99, NSL-KDD, etc	Varies; up to 99.4%
Elmrabit et al. (2020), Jaiswal et al.(2020), Diaba and Elmusrati (2023), Siniosoglou et al. (2021), Mukherjee et al. (2022)	Equipment Malfunctions and Faults	DT, ARIMA, LSTM, etc	Real-world and simulated datasets	Varies; up to near-perfect accuracy
ali et al. (2022), Fenza et al. (2019), Hashatsi et al. (2020), Madhure et al. (2020), Vijayanand et al. (2019)	Energy Consumption Irregularities	CNN-LSTM, FL, etc	Real-time smart meters' data, public datasets, etc	Varies; up to 97.84%
Esmalifalak et al. (2017), Danilczyk et al. (2020), Qi et al. (2023), Qi et al. (2021), Martinelli et al. (2022)	Anomalies in Energy Distribution and Demand Forecasting	FL, Unsupervised ML Techniques, etc	Open data sets, trial studies, etc	Varies; high accuracy reported

izing models across different environments and assembling large, labeled datasets for training, which can be time-consuming and costly. Furthermore, certain models have reported challenges in detecting more subtle anomalies or in environments with significant noise and data imbalance (Ali et al. 2022; Madhure et al. 2020; Mukherjee 2023). Therefore, there is a need for further research into models that can effectively learn from smaller, more diverse,

or less balanced datasets and operate robustly under a variety of operational conditions. The deployment of these models in real-world settings should be explored in future studies, considering dynamic grid conditions and different attack vectors (Kawoosa and Prashae 2022; Menon and Radhika 2016). It remains a challenge to ensure that models are scalable and adaptable to different grid sizes and configurations. There is a potential for developing systems that integrate anomaly detection with existing grid management systems for real-time monitoring and response (Karimipour et al. 2019; Oprea et al. 2021). Moreover, models that can update their parameters in real-time through continuous learning may be able to adapt more effectively to evolving grid conditions and emerging threats. There is a need for further research into ensuring the privacy and security of shared data for collaborative learning models, as FL approaches show promise (Jithish et al. 2023). It is particularly relevant since SGs rely heavily on critical infrastructure with significant security implications. A comparative study of various anomaly detection techniques and establishing benchmarks for SG anomaly detection would provide utility providers with clearer guidance on the most appropriate approach to take (Elmrabit et al. 2020; Danilczyk et al. 2020; Dai et al. 2020). While these results are satisfactory, further research, as discussed in Sect. 7, could explore specific limitations identified in each paper.

7 Future research directions

After conducting a thorough review, we have proposed avenues for future research in anomaly detection within smart grid systems using AI techniques. These directions cover crucial areas, including foundation models, edge intelligence, privacy-preserving mechanisms, Cybersecurity resilience, data quality, real-time detection, explainability, and regulatory frameworks. Addressing these challenges is vital for ensuring that AI-powered anomaly detection systems are accurate, scalable, secure, transparent, and ethically aligned with the critical demands of energy infrastructure. As smart grids become increasingly complex and interconnected, the ability to robustly detect anomalies in real time, while preserving user privacy and system integrity, is essential. Without progress in these research areas, the deployment of AI in smart grids may face limitations due to poor generalizability, low operator trust, or susceptibility to adversarial threats. Table 10 provides a concise overview of the future research directions, highlighting key gaps and the proposed approaches to address them.

8 Conclusion

As SG technology continues to advance, extensive research has been conducted on the application of ML techniques for anomaly detection. This research highlights the crucial role of ML in addressing the numerous challenges faced by modern power systems. SGs, with their integration of diverse and renewable energy sources, have brought about significant changes in the power landscape. These changes have introduced complexities that require innovative solutions to maintain grid integrity and reliability. Our comprehensive review has identified several anomalies associated with SGs, including electricity theft, cyber-attacks, power system disturbances, and abnormal consumption patterns. In this study, we systematically

Table 10 Future research directions

Theme	Research gaps	Proposed approaches
Foundation Models & Cross-Domain Learning	1. Lack of pre-trained models for grid-specific anomaly tasks 2. Poor generalization across regions or grid types 3. Sparse labeled data	1. Fine-tune foundation models (e.g., PowerGPT) for grid tasks 2. Use transfer/self-supervised learning to adapt across regions 3. Integrate multi-modal data (e.g., sensor, asset, text) via LLMs
Edge AI & Neuromorphic Computing	1. High latency in centralized models 2. Limited compute on edge devices 3. Underexplored neuromorphic systems	1. Deploy spiking neural networks on neuromorphic chips 2. Use lightweight models for RTUs/PMUs 3. Decentralized/federated edge training with local explainability
Federated & Privacy-Preserving Detection	1. Privacy issues with smart meter data 2. Bottlenecks in central FL servers 3. Lack of privacy-compliant frameworks	1. Implement differential privacy, secure FL, or gossip-based P2P FL 2. Use blockchain or SMPC for data traceability and audit trails
Electricity Theft & Energy Fraud	1. Class imbalance 2. Adaptive fraud techniques 3. Data silo limitations	1. Use DL models (CNN/LSTM/transformer) with data augmentation 2. Enable collaborative federated detection across grids 3. Graph-based models for collusive theft detection
Cybersecurity & Attack Resilience	1. High false positives in current systems 2. Evolving cyber threats (FDIAs, malware) 3. AI model vulnerability to adversarial attacks	1. Combine rule-based + AI models for layered detection 2. Apply adversarial training, threat intelligence, and explainable AI 3. Use LLMs for contextual alerts (e.g., APT pattern explanation)
Data Quality & Integrity	1. Sensor noise, missing data, injection attacks 2. Lack of robust cross-check mechanisms	1. Use digital twins, robust ML, and autoencoders for imputation 1. Cross-validate across SCADA, weather, and metering layers
Spatio-Temporal & Physical Fault Detection	1. Complex spatial grid dynamics 2. Nonstationarity due to renewables 3. Lack of labels for rare faults	1. Employ spatio-temporal GNNs and physics-informed models 2. Simulate anomalies via digital twins 3. Use neuromorphic event-driven sensors for transient faults
Predictive Maintenance & Prognostics	1. Sparse failure data 2. Diverse asset types 3. Unused textual and image data	1. Use time-series transformers and multimodal models 2. Combine inspection logs, drone images, and sensors 3. Transfer learning across asset types
Explainability, Transparency & Fairness	1. Black-box models 2. Biased training data 3. No standard explainability metrics	1. Apply SHAP/LIME, counterfactuals, and user-friendly visualizations 2. Audit for fairness across consumer groups and integrate bias mitigation

classify these anomalies and evaluate various ML models, ranging from supervised to unsupervised to semi-supervised and RL, in terms of their ability to detect and mitigate these anomalies. Each model is thoroughly analyzed, outlining its nuanced capabilities and limitations, providing a clear understanding of how it can be applied to different SG scenarios.

A crucial finding of our research is that ML holds significant promise for enhancing grid security and operational efficiency. However, our study also identifies formidable challenges, such as data availability, model interpretability, and the adaptability of these models

to the ever-changing dynamics of the grid. It is important to note that these challenges are not solely technical in nature but also involve regulatory, privacy, and ethical considerations, particularly with regard to consumer data security and system operation. We suggest a multidisciplinary approach to address these challenges, advocating for collaboration among power system engineers, data scientists, and policymakers. Such collaboration is essential to develop technically sound, ethically and legally compliant ML models. Furthermore, our paper emphasizes the significance of continuous research and development to keep pace with the rapidly evolving nature of SGs and the associated cyber threats. We have proposed ten future research direction themes to assist future researchers.

Overall, ML shows great promise in detecting anomalies in SGs. However, this journey is fraught with challenges that require concerted efforts and innovative solutions. Future research directions include improving data quality and availability, enhancing model interpretability, and developing models that are resilient to new types of anomalies and cyber threats. Despite the ever-changing energy landscape, the ultimate goal remains clear: to grasp the power of ML to foster SGs that are efficient and also reliable, secure, and resilient. While our review focused on AI techniques for anomaly detection in SGs, we acknowledge the relevance of emerging technologies such as edge computing, quantum ML, and blockchain. In future work, we plan to explore their potential to enhance AI-based anomaly detection.

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